Less Simple, Less Distortionary?

Evidence from Costa Rica's Corporate Tax Reform

Lucía Contreras[†] Jonathan Garita[‡*]

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Abstract

Simple but distortionary tax designs are widely used in the taxation of firms in developing economies. This paper examines whether less simple designs, which can reduce these distortions, introduce new ones as firms face less simple tax incentives. We study Costa Rica's 2020 Corporate Income Tax reform, which replaced a simple but highly distortionary notched schedule with a less simple kinked one. Using the variation in tax rates it created across firms, we show that firms react to the right incentives by adjusting their reported profits to changes in marginal tax rates, not in average tax rates, without the average-rate mistakes documented for individuals. Responses to tax cuts and increases are also symmetric, with elasticities of reported profits of about 3.1–3.2, suggesting that the concern raised for other taxes, that reversing a tax change may not undo its effects, may be smaller for corporate taxes. Bunching below the replaced notch and cost inflation above it also disappeared, and little or no bunching emerged at the new profit thresholds. We argue that less simple tax designs can reduce distortions without firms misreading their incentives, and should hence be considered in developing countries as a more economically efficient alternative.

Keywords: corporate income tax; tax simplicity; notches and kinks; marginal and average tax rates; developing countries

JEL Codes: H25, H21, H26, D22, D91, O23

*†University of Manchester, lucia.contreras@manchester.ac.uk. ‡Central Bank of Costa Rica. Lucía is grateful to Mazhar Waseem and James Banks for their guidance and encouragement. We thank seminar and conference participants at the University of Costa Rica, the Central Bank of Costa Rica, the University of Manchester, Universidad de los Andes, the NWSSDTP Conference, the Royal Economic Society Doctoral and Annual Conferences, the Institute for Fiscal Studies, the ZEW Public Finance Conference, the OECD Tax Policy and Statistics Division, the Mannheim Taxation Conference and the Oxford CBT Doctoral Conference for their comments and suggestions. We are grateful to the Central Bank of Costa Rica for providing access to the data. The views expressed herein are those of the authors and do not necessarily represent the views of the Central Bank of Costa Rica. All results have been reviewed by the Central Bank of Costa Rica to ensure that no confidential information is disclosed. All errors are our own.

1 Introduction

Developing economies often rely on simple tax designs to apply lower rates to smaller firms. Their appeal comes from reducing the information firms need to determine their tax obligations, lowering compliance costs and making tax rules easier to understand (Slemrod, 2013; Aghion et al., 2025). However, these simple rules can also generate strong behavioural distortions when tax liabilities change sharply at cutoffs, as with eligibility thresholds for simplified tax regimes for small firms (Mas-Montserrat et al., 2023; Bashir et al., 2024) and notched schedules (Kleven and Waseem, 2013). Reducing such distortions may require less simple tax schedules. Evidence from individuals suggests that, under these, taxpayers respond to the wrong incentives or fail to adjust their behaviour as intended. When faced with relatively complex non-linear tax or pricing schedules, individuals often deviate from what standard models of optimisation would predict, responding to the average tax rate or price rather than to the relevant marginal one when making work or consumption decisions, a heuristic known as *ironing* (Liebman and Zeckhauser, 2004; de Bartolomé, 1995; Ito, 2014; Rees-Jones and Taubinsky, 2020; Shaffer, 2020; Ho, 2023), and adding complexity reduces people's responsiveness to new incentives, even when those incentives are clearly communicated (Abeler and Jäger, 2015). In this paper, we find no evidence of such mistakes and behavioural biases among firms in Costa Rica, and hence argue that less simple, and more economically efficient, tax designs should be considered in developing countries.

The degree to which incorporated firms, in particular in developing countries, can correctly identify and respond to the relevant incentives under less simple tax schedules, without introducing new distortions, remains largely unknown. There are reasons to expect this to be harder for firms in developing economies, which are typically smaller (Hsieh and Olken, 2014), often keep limited records (Tourek, 2022), make reporting errors that increase their tax liability (Almunia et al., 2024), and operate alongside a large informal sector (La Porta and Shleifer, 2014). To fill this gap, we provide new evidence by analysing Costa Rica's 2020 Corporate Income Tax (CIT) reform, which changed the design of the CIT in several ways, most notably by replacing a simple but highly distortionary notched schedule with a less simple kinked one. Under a notched schedule, a single average rate applies to the entire tax base, making tax liabilities easy to compute for both firms and revenue authorities. Yet crossing a cutoff subjects the whole tax base to a higher rate, causing tax liability to increase discontinuously and generating strong incentives to remain below it (Kleven and Waseem, 2013; Bachas and Soto, 2021). Kinked schedules, common in developed countries, are less simple but do not have this feature.

Prior to the reform we study, Costa Rica applied a notched design by taxing firms' profits

at a rate determined by two thresholds of reported revenue. Firms bunched just below the revenue thresholds and sharply increased reported costs above them to reduce taxable profits (Bachas and Soto, 2021). These behavioural responses, a clear sign of the distortions created by the notches, were among the concerns that motivated policy makers to introduce the reform. For small and medium firms, the new schedule replaced the lower revenue threshold with a three-threshold kinked schedule in which the tax rate depends on reported profits rather than revenue, while the higher revenue threshold and the single average tax rate for large firms remained unchanged. Unlike notches, kinks apply higher marginal rates only to the part of the tax base above each threshold (Saez, 2010; Chetty et al., 2011), reducing the distortion at the cutoff but requiring firms to account for different rates applying to different portions of their profits, and to distinguish between marginal, relevant for intensive margin decisions, and average rates. Small and medium firms therefore faced several changes at once: a new variable determining their rate (from revenue to profits), new thresholds and rates, and a schedule in which marginal and average rates no longer coincide, which made ironing possible for the first time and thus could lead them to respond to the wrong incentives. This reform thus provides a setting to examine how firms respond to the incentives of a less simple tax schedule.

We use administrative corporate tax returns from 2016 to 2024 to track firms before and after the reform. The redesign generated substantial heterogeneity in tax changes: for a given level of revenue and profits, firms could face increases, decreases, or no change in their marginal tax rate, while their average tax rate could move by a different amount or in a different direction. We exploit this variation in a difference-in-differences design, comparing small and medium firms exposed to different reform-induced changes in marginal and average tax rates with larger firms that experienced neither, our control group. Because post-reform tax rates depend on firms' reported outcomes and are therefore endogenous to their behavioural responses, we assign treatment using the tax changes firms would have faced had their reported revenue and profits remained at their pre-reform levels. This allows us to estimate responses to marginal and average tax incentives. Our main outcome is reported profits, and we also examine reported revenues and costs to identify the margins through which firms adjust.

We show that Costa Rican firms respond to marginal rather than average tax rates when adjusting outcomes at the intensive margin. Reported profits adjust rapidly to reform-induced changes in marginal tax rates, while we find no evidence that variation in the average tax rate affects reported profits. In particular, firms for which the marginal tax rate changes adjust their reported profits in the direction predicted by standard after-tax profit maximisation, including when marginal and average tax rates move in opposite directions,

where ironing and the marginal incentive predict opposite responses. Firms whose marginal tax rate remains unchanged but whose average tax rate falls provide a more direct test of responses to average rates: for one such group, reported profits do not change significantly, while for the other they fall, the opposite of the response that a lower average tax rate would entail. Despite facing a new schedule with several rates, where marginal and average tax rates no longer coincide, and multiple changes introduced at the same time, firms in Costa Rica therefore appear able to respond promptly to the relevant incentives without the average-rate behavioural biases documented for individuals and consumers.

A second finding is that firms respond symmetrically to changes in marginal tax rates. Firms facing a lower marginal rate increase their reported profits, and firms facing a higher rate reduce them, with very similar elasticities. The elasticity of reported profits with respect to the marginal net-of-tax rate is 3.13 for firms whose rate fell and 3.18 for firms whose rate rose, a difference that is small relative to the precision with which each is estimated. This contrasts with the asymmetric responses documented for other taxes, and suggests that the same elasticity can be used to assess corporate tax cuts and tax increases.

While the magnitude of the profit response is similar for marginal tax decreases and increases, the margins of adjustment differ. Reported revenue increases for firms facing tax changes in either direction, consistent with the replacement of the lower revenue notch, which had held back reported revenue growth. Reported costs also increase in both groups, but by considerably more for firms facing higher marginal rates. As a result, revenue increases by more than costs when marginal rates fall, while costs increase by more than revenue when marginal rates rise. The stronger cost response among firms facing higher rates is consistent with Bachas and Soto (2021), who show that reported costs are an important margin of tax adjustment in this setting.

Finally, changes in reporting around the tax thresholds provide complementary evidence that the reform reduced the distortions at the lower revenue threshold, where the notch was replaced, with little sign of bunching at the new profit thresholds. After the reform, the excess and missing mass around the lower revenue threshold disappeared, as did the drop in profit margins and the jump in reported costs above it. In contrast, bunching remained at the higher revenue threshold, where the notch stayed in place, and the discontinuities in profit margins and reported costs became more pronounced there, consistent with the reform increasing the jump in tax rates at this threshold. Average employment remained smooth around both revenue thresholds before and after the reform, suggesting that the responses around these thresholds may in good part reflect reporting rather than real adjustments. At the three new profit thresholds introduced by the kinked schedule, the distribution of reported profits shows very little or no excess mass.

This paper contributes to several strands of literature. First, it adds to the literature on behavioural responses to non-linear tax and pricing schedules. Existing evidence for individuals and consumers documents misoptimisation under kinked schedules, notably ironing: responding to average incentives, rather than to marginal incentives relevant for making decisions on consumption and work at the intensive margin (Liebman and Zeckhauser, 2004; de Bartolomé, 1995; Ito, 2014; Rees-Jones and Taubinsky, 2020; Shaffer, 2020; Ho, 2023). Whether these findings extend to firms is not clear a priori. Firms may be better equipped to respond to tax incentives, with evidence that managerial ability influences corporate tax decisions (Koester et al., 2017) and that tax preparers can help firms avoid optimisation errors (Zwick, 2021). At the same time, surveys in the accounting literature find that average tax rates are often used in US corporate investment and financing decisions (Graham et al., 2017), and evidence from low- and middle-income countries shows that firms can make costly reporting mistakes that raise their tax liability (Almunia et al., 2024; Tourek, 2022). Using administrative tax returns, we provide, to our knowledge, the first evidence from firms' actual tax reporting that they distinguish marginal from average rates under a kinked corporate tax schedule.

Second, the paper contributes to the emerging literature on corporate taxation in developing countries. Existing work documents large responses to corporate taxes in settings where limited enforcement creates greater scope for reporting adjustments (Alejos, 2018; Bachas and Soto, 2021; Waseem, 2018; Lobel et al., 2024). Much of this evidence relies on bunching around tax thresholds and therefore identifies local responses to tax incentives. By exploiting tax rate changes induced by the reform across a broad range of firms, we provide a non-local estimate of the elasticity of reported corporate profits. Our preferred estimates imply an elasticity of reported profits with respect to the marginal net-of-tax rate of between 3.13 and 3.18. This is large relative to estimates for developed countries, which are typically below 1 (Gruber and Rauh, 2007; Dwenger and Steiner, 2012; Devereux et al., 2014; Patel et al., 2017), but in line with the elasticities of between 2.7 and 4.8 estimated for Costa Rica by Bachas and Soto (2021) and with the large reporting responses to corporate taxes documented in other developing economies (Alejos, 2018; Lobel et al., 2024).

Third, we contribute to the literature on whether responses to changing tax incentives differ between tax increases and decreases. Asymmetries between tax increases and decreases have been documented for VAT, payroll taxes, individual income taxes and estate taxes (Benzarti et al., 2020; Saez et al., 2019, 2021; Hussain and Malik, 2016; Wu and Yang, 2024). These asymmetries matter for policy in two ways. First, the effects of a tax change depend on its direction, so an elasticity estimated from tax cuts cannot be used to inform the effects of tax increases, or the reverse. Second, reversing a tax change may neither undo its effects

nor return the economy to its previous state: prices respond much more to VAT increases than to VAT cuts, so a cut followed by an increase of the same size can leave prices higher than before (Benzarti et al., 2020), and the employment gains from a payroll tax cut for young workers in Sweden persisted after the cut was reversed (Saez et al., 2021). Identifying such asymmetries typically requires comparing reforms that occurred at different times, so estimated differences could also partly reflect the circumstances of each episode rather than the direction of the tax change itself. Costa Rica's reform raised marginal rates for some firms and lowered them for others within a single policy event, allowing the comparison to be made directly. We find responses of similar magnitude in both directions, suggesting that these asymmetries may not be a general feature of responses to taxation. For the corporate income tax, a single elasticity could therefore guide tax changes in either direction.

The remainder of the paper is organised as follows. Section 2 describes the data. Section 3 presents Costa Rica's corporate tax system and the 2020 reform. Section 4 discusses the responses we expect to changes in marginal and average tax rates, and what ironing would imply for firms. Section 5 describes the empirical strategy. Section 6 presents the results. It first shows that reported profits responded to changes in the MTR in the expected direction, with similar magnitudes for tax cuts and increases. It then examines identification, robustness and the margins of adjustment, and finally asks whether firms responded to the MTR or to the ATR. Section 7 concludes.

2 Data

We use the 2016–2024 Registry of Economic Variables (Revec), a dataset compiled by the Central Bank of Costa Rica. This dataset contains the complete set of administrative corporate tax returns submitted to the Ministry of Finance. All registered and formal corporations in Costa Rica are required to file an electronic annual tax declaration (D-101), which includes typical balance sheet information and tax reports such as reported revenues, costs, and profits. These data are used to calculate corporate taxes based on the prevailing tax structure, as described in Section 3. In addition to tax returns, the Central Bank of Costa Rica enhances the dataset with firm-level information, such as employment, wage bill and industry classification. These supplementary data are sourced from administrative records, including social security and the national firm registry. We also calculate additional variables like the profit margin (defined as reported profits over reported revenues) using this dataset.

For this study, we focus on firms with records of reported revenue and costs whose maximum pre-reform annual revenue is below 200,000,000 CRC (approximately 372,000 USD

in 2014[1]), normalised relative to the second reported revenue threshold established in 2014.[2] We exclude firms that never report annual revenues exceeding 3,000,000 CRC (approximately 5,600 USD) before the reform, to eliminate those with minimal activity, and keep firms registered as active, with positive reported revenue and costs. This gives a sample of 49,827 firms and 348,340 firm-year observations. Finally, we winsorise reported revenue, costs and profits at the 1st and 99th percentiles within each year.

Our main sample imposes two further restrictions. First, we keep firms observed in every year from 2016 to 2024. Since we study intensive margin responses, a balanced panel ensures that the estimates reflect changes in the reports of the same firms, not changes in which firms enter or leave the sample. Second, we keep firms with formal employees reported to the Social Security Authority in at least one pre-reform year. This restriction allows us to control for firm size and labour intensity, through pre-reform employment and the ratio of the wage bill to revenue, which makes treated and control firms more comparable. It also excludes firms with no formal employees. These firms are smaller, report higher profit margins, are concentrated in professional services, and include almost all firms with no industry recorded (99% of the 1,644 firms without a recorded industry), for which we cannot precisely control for industry-specific shocks. The main sample comprises 12,668 firms and 114,012 firm-year observations.

Table 1 compares the main sample using a balanced panel with firms with employment records (Column 4) with less restricted samples. Each restriction keeps somewhat larger firms: average pre-reform revenue rises from 48.6 million CRC among all firms to 68.1 million CRC in the main sample, and the share of large firms, which form our control group (Section 5.1), from 11% to 19%. Profit margins, however, are similar across samples, between 0.12 and 0.14. We also present results using the unbalanced panels, with and without the employment restriction.

[Table 1 here]

[1]Values in USD are converted using the 2014 average official Central Bank's exchange rate of 538.3 CRC per USD.

[2]The thresholds are updated annually and closely follow the previous year's consumer price inflation, with a correlation of 0.98 between their annual changes over 2006–2024. We therefore re-centre the data by normalising each year's observations to the second reported revenue threshold of 2014, ensuring that all thresholds remain constant over time. This transformation allows us to consistently compare bunching behaviour across different years. We use this normalised data for data cleaning and to produce descriptive graphs.

3 Institutional Setting

Costa Rica enacted a major corporate tax reform with the approval of the Law on the Strengthening of Public Finances by Congress on December 3, 2018.[3] The CIT reform was implemented on October 1, 2019, marking the start of the 2020 fiscal year.[4] One of its aims was to reduce the distortions created by the notches of the previous design, which gave firms clear incentives to bunch below the revenue thresholds and to inflate their reported costs above them.

Prior to the 2020 fiscal year, Costa Rica imposed an increasing average tax rate (ATR) on reported profits, defined as reported revenue minus deductible costs. While the statutory rate was 30%, smaller firms benefited from two preferential lower tax rates based on their reported revenues. As illustrated in Figure 1, both the ATR and marginal tax rate (MTR) exhibited sharp jumps around two reported revenue thresholds under a notched tax schedule. When a firm's reported revenue crossed a threshold, the higher rate applied to all reported profits, not just the portion above the threshold. At the first reported revenue threshold (T_1, equal to annual reported revenues of 49,969,000 CRC or approximately 93,000 USD in 2014), the ATR jumped from 10% for small firms to 20% for medium firms. At the second threshold (T_2, equal to 100,513,000 CRC or approximately 187,000 USD in 2014), the ATR increased further from 20% to the statutory 30% rate for large firms. Under this design, the determinant of the tax rate (reported revenue) differed from that of the tax base (reported profits), and the ATR was always equal to the MTR.

[Figure 1 here]

The design of notched tax systems, as analysed by Kleven and Waseem (2013), Londoño-Vélez and Avila-Mahecha (2024), and other related literature, is known to be highly distortionary. Firms subject to these systems face strong incentives to relocate from just above the threshold to just below it in order to qualify for a lower tax rate. In the case of the CIT system in place before the reform, this behaviour led to a clear excess mass of firms bunching just below the thresholds, as well as signs of missing mass just above them. Figure

[3] The reform package also replaced the sales tax with a value-added tax, established a fiscal rule to limit public spending, and restructured the salary framework for public employees.

[4] Costa Rica also redefined its fiscal year. Before the reform, the fiscal year spanned from October to September. With the reform, the fiscal year changed to align with the calendar year, running from January to December, and starting from the 2021 fiscal year onwards. The 2020 fiscal year was a transitional year, extended for 15 months from October 2019 to December 2020. Following the Central Bank of Costa Rica, which reports 2020 revenue as a 12-month equivalent, we scale all 2020 tax return values proportionally to 12 months.

2A illustrates this, showing the firm density for three years preceding the reform (2016, 2017 and 2018). Tax reports also show a sharp drop in reported profit margins at both thresholds (Figure 2B). Average profit margins follow a downward step pattern: they are roughly constant within each bracket and fall discontinuously at each threshold. This is consistent with firms just above a threshold reducing reported revenue to stay below it, and with firms already above it reporting higher costs to reduce their taxable profits. Bachas and Soto (2021) show that these cost discontinuities are too large to be explained by real production responses, since employment, wage bills and capital remain continuous at the thresholds, and conclude that the cost responses largely reflect tax evasion, while part of the revenue responses also reflects misreporting.

[Figure 2 here]

3.1 CIT After the Reform

After the 2020 CIT reform, large taxpayers, those with reported revenue exceeding the second reported revenue threshold (T_2), continue to face a statutory ATR of 30% on reported profits. For medium and small firms, the tax schedule changed in two major ways. First, the variable determining their preferential tax rates changed: these rates are now determined by reported profits, not revenue. Second, the schedule was redefined with new thresholds and new MTRs: the first revenue threshold (T_1) was replaced by three profit thresholds, so small and medium firms now face the same schedule. Instead of a single rate on all their profits, these firms now apply up to four rates to different portions of them. As a consequence of this, marginal and average tax rates no longer coincide. Under the new schedule with kinks, higher MTRs apply only to the part of profits above the corresponding thresholds, so the ATR increases smoothly with profits and, above the first profit threshold, is lower than the MTR.

These changes are summarised in Figure 3. Panel A shows that the MTR for small and medium-sized firms ranges from 5% to 20%, determined by three new profit thresholds: t_1 (equal to 4,741,000 CRC or approximately 8,800 USD), t_2 (7,112,000 CRC or approximately 13,200 USD), and t_3 (9,482,000 CRC or approximately 17,600 USD).[5] Panel B shows that for large firms, the MTR and the ATR remained at 30% after the reform. The reform thus replaced the notch at the first revenue threshold with a kinked schedule, but retained the notch at the second, above which large firms still pay the statutory 30% rate on all their profits. Table 2 summarises the main features of the CIT before and after the reform.

[5] Values expressed in 2014 CRC and USD.

[Table 2 here]

[Figure 3 here]

Descriptive analysis reveals large and rapid changes in reporting behaviour in the years following the reform, 2021, 2022 and 2023. First, Figure 4A shows the disappearance of excess and missing mass around the replaced first notch, consistent with the removal of incentives to bunch at that point. Second, bunching remains evident at the second threshold, which still separates small and medium firms from large ones. The incentives to bunch at this upper reported revenue threshold persist and have even increased, as the MTR and ATR before this threshold decreased for many firms. Specifically, before the reform, medium firms crossing the second threshold experienced a jump of 10pp in the rate applied to all their profits, from 20% to 30%. After the reform, crossing this threshold can now result in a jump of up to 25pp in that rate for lower profitability firms.

Third, we observe a decrease in firm density for the bins below the first notch and an increase for the bins above it, suggesting that the first notch may have disincentivised firms from growing —or at least from reporting higher revenues to the tax administration. This trend becomes clearer when examining the density changes around the first threshold, as depicted in Figure 5A. Here, we observe that with the reform, firm density decreases before and increases after this replaced threshold. However, we do not observe a similar pattern around the second threshold that remained in place, as seen in Figure 5B. Instead, we see a general increase in firm density in that vicinity.

[Figure 4 here]

[Figure 5 here]

The reform also led to changes in the profit margin behaviour pattern, as seen in Figure 4B. The previously observed discontinuity in profit margins at the first threshold vanishes, leading to noticeably higher average profit margins for medium firms. Meanwhile, the profit margin discontinuity around the still-present second threshold becomes more pronounced, consistent with the higher incentives to lower profit margins when crossing the threshold. The role of cost declarations in this pattern becomes clear in Figure 4C. After the reform, the jump in cost declarations smooths out at the first threshold and becomes sharper at the second one. All these descriptive patterns suggest prompt behavioural responses to the CIT

reform. Average employment, depicted in Figure 4D, grows steadily with reported revenue before and after the reform, and so does its dispersion, with no discontinuity near either of the two thresholds. This suggests that the discontinuities observed in density, reported profit margins, and reported costs around the thresholds may in good part reflect reporting rather than real adjustments.

As the corporate tax rate of small and medium firms started to be determined by profits, we also examine whether firms bunched at the new profit thresholds. Figure 6 presents the distribution of reported profits of small and medium firms before and after the reform. This shows, at most, a small excess mass just below t_1, much less marked than the bunching at the revenue notches before the reform, and no visible excess mass at t_2 or t_3. This suggests that the kinks have not introduced strong incentives to bunch at these thresholds, at least in the post-reform years we observe, which is consistent with kinks creating weaker incentives to bunch than notches.

[Figure 6 here]

3.2 Variation in Tax Rates

By altering the design of the tax schedule, the reform introduced heterogeneous tax rate changes across firms. For small and medium firms, the rate became a function of reported profits instead of revenue, and the first notch was replaced by several kinks with new rates. As a result, firms' MTR and ATR changed by different amounts and in different directions, depending on their size and reported profits. For instance, a firm could face a change in its MTR from 10% or 20%, to either 5%, 10%, 15%, or 20%, for the same tax report. Specifically, and as shown in Figure 7, small firms with reported revenue up to the first reported revenue threshold (T_1) saw a decrease, no change, or an increase in their MTR. For medium firms with reported revenue between T_1 and T_2, the MTR either decreased or remained unchanged. Large firms did not experience a change in their MTR. The ATR, in turn, fell for medium firms at all profit levels, while for small firms it fell below P (the calculated level of profits at which the change in the average tax rate of small firms turns from negative to positive) and rose above it.

Figure 7 also shows how we split firms into seven groups, by firm size and reported profits, depending on how the reform changed their MTR and ATR for an unchanged tax report. Both rates fell for G1 and G5 and both rose for G4, by different amounts. The two rates moved in opposite directions for G3, whose MTR rose while its ATR fell, and only the ATR

changed for G2 and G6. These six groups are our treatment groups, and large firms (GC), whose rates did not change, are the control group. Section 5.1 explains how we exploit these groups in our empirical strategy.

[Figure 7 here]

4 Conceptual Framework

As a benchmark, consider the standard model in which firms maximise after-tax profits. Here, intensive margin decisions, such as the level of production or the amount of profit reported to the tax authority, depend on the MTR, which sets the tax due on an additional unit of profit and hence the after-tax return to each marginal adjustment. Extensive margin or discrete decisions, such as entry and exit or the location of investment, instead depend on the ATR, which determines total after-tax profits under each alternative (Devereux and Griffith, 2003). Our analysis focuses on the intensive margin, mainly on reported profits and on the revenues and costs through which they adjust.

For reported profits, the model predicts that a reduction in the MTR raises them and an increase lowers them. Two channels operate in the same direction in a low enforcement capacity setting, such as Costa Rica's (Bachas and Soto, 2021). On the real side, a lower MTR raises the after-tax return to an additional unit sold, which encourages firms to expand production. On the evasion side, a lower MTR reduces the tax saved by each unit of revenue under-reported or cost overstated, and with it the incentive to evade. An increase in the MTR works in reverse through both channels, reducing real activity and strengthening the incentive to under-report revenues or inflate costs. In a frictionless setting, the response is determined by the elasticity of reported profits with respect to the net-of-tax rate, $1 - MTR$, so tax cuts and tax increases of similar size should lead to responses of similar magnitude.

We are interested in whether firms depart from this benchmark and respond to the ATR instead, as individuals have been shown to do under kinked schedules. When the MTR differs from the ATR, consumers and individual taxpayers frequently *iron* the schedule: they use their ATR to make decisions at the intensive margin and not the MTR (Liebman and Zeckhauser, 2004). The rationalisation of this behaviour is that the average rate is more salient. Taxpayers observe or compute their total tax liability, whether from a past return or from the current schedule, and the average rate follows directly from it. Finding the marginal rate requires identifying the bracket in which the last unit of income falls and the rate that applies to it (Liebman and Zeckhauser, 2004; Rees-Jones and Taubinsky, 2020).

For example, a taxpayer who works out the total tax due on their expected income may assume that any extra earnings will be taxed at the same average rate, without checking the higher bracket that applies to them.

Among individuals, this behaviour has been documented in experiments in which participants make choices under a given tax schedule (de Bartolomé, 1995), in how taxpayers forecast the tax due on additional income (Rees-Jones and Taubinsky, 2020), and in household responses to kinked electricity prices (Ito, 2014; Shaffer, 2020). It is attributed to taxpayers simplifying a tax or price structure they do not fully understand. Importantly, ironing has implications for the design of kinked schedules. If taxpayers iron, their responses depend on average rather than marginal rates, so a reform that changes marginal rates may not affect behaviour and revenue as standard models predict (Liebman and Zeckhauser, 2004; Rees-Jones and Taubinsky, 2020; Ho, 2023). For firms, ironing would imply that reported profits respond to the ATR rather than to the MTR, and hence away from after-tax profit maximising behaviour. Whether firms make the same mistake is not clear a priori. They may be better placed than individuals to identify their marginal rate, for example by relying on professional tax advice. In developing economies, however, this may be harder, since firms are typically small (Hsieh and Olken, 2014), often keep limited records (Tourek, 2022), and can make reporting errors that raise their tax liability (Almunia et al., 2024). In our setting, the reform also changed several features of the schedule at once, and firms now apply up to four rates to different portions of their profits. To our knowledge, whether firms iron the tax schedule has not yet been tested.

The benchmark yields three predictions, which we test in two steps. The first step uses only the variation in the MTR, comparing firms whose MTR fell or rose with large firms, whose rates did not change (Sections 6.1 to 6.3). It tests the first two predictions. First, reported profits should increase for firms whose MTR fell and decrease for firms whose MTR rose. Second, absent frictions, responses to tax cuts and tax increases should be of similar magnitude. The second step tests the third prediction (Section 6.4). Reported profits should respond to the MTR and not to the ATR, whereas firms that iron the schedule would respond to the ATR.

The second step requires comparing groups that faced different combinations of changes in the two rates. The groups defined in Section 3.2 provide these combinations. They differ in the direction of the changes in their two rates, and in two of them only the ATR changed. Under the benchmark, reported profits should rise for groups whose MTR fell (G1 and G5), fall for groups whose MTR rose (G3 and G4), and remain unchanged for groups whose MTR did not change (G2 and G6). For G1, G5 and G4, whose two rates moved in the same direction, ironing and a response to the MTR predict the same change in reported profits.

For G3, whose MTR rose while its ATR fell, and for G2 and G6, whose ATR fell while their MTR did not change, ironing would instead lead to higher reported profits. The test of ironing therefore rests on these three groups. The predicted effects on reported profits for the different changes in MTR and ATR experienced by the seven groups are summarised in Table 3.

[Table 3 here]

5 Empirical Strategy

This section outlines how we use the variation in tax rates induced by the reform to investigate firms' behavioural responses to marginal and average corporate tax rates, and to test the predictions set out in Section 4. We employ difference-in-differences (DiD) research designs, comparing small and medium firms whose tax rates changed with large firms, whose rates did not, before and after the reform. We first describe how we define treatment, based on the changes in tax rates that firms would have faced had they reported the same revenue and profits as before the reform, and then present our estimations. Our first estimates use changes in the MTR, both as a binary treatment and through the net-of-tax rate, and we then compare groups that faced different combinations of changes in the MTR and ATR.

As we focus on evaluating intensive margin responses to changes in tax rates, our main specifications use the balanced panel of firms with employment records described in Section 2 (Table 1, Column 4). We also compare our baseline results with those using the unbalanced panel of firms with employment records.

5.1 Treatment Definition

We estimate firms' responses to changes in tax rates by exploiting the variation in marginal and average tax rates introduced by the reform. Directly using the actual changes in tax rates to define binary treatments, or the actual net-of-tax rates for elasticity estimates, would lead to biased estimates, as these are endogenous to firms' responses: after the reform, the rates firms face depend on the revenue and profits they report, which are our outcomes. To address this, we follow the established literature and use the *mechanical tax rate changes* instead (Weber, 2014). The mechanical tax change is defined as the percentage point variation in the tax rate (MTR or ATR) that a firm would face after the reform, assuming it continued to report the same tax information as before the reform, specifically the same reported revenue

and profits. It therefore captures what would have happened to each firm's tax rates in the absence of any response, and uses only pre-reform information to define treatment groups and to construct a measure of each firm's exposure to the tax changes introduced by the CIT reform.

Because the mechanical changes depend on firms' pre-reform revenue and profits, they may also be correlated with transitory shocks to those reports, which would later revert to the mean. To mitigate mean reversion concerns, as well as potential anticipation of the reform and firms' self-selection into the treatment or control groups, we construct the mechanical tax rate changes using the average of firms' 2016–2019 tax reports, instead of relying solely on the single-year 2019 pre-reform tax declaration. Averaging over four years smooths out transitory shocks to any single year.

We define treated firms as those that experienced changes in their mechanical tax rates. The control group (GC) consists of large firms, whose mechanical marginal and average tax rates did not change. For example, a firm reporting an average of CRC 45 million in revenue and CRC 4 million in profits during the 2016–2019 pre-reform period would experience a change in its MTR from 10% to 5%, based on the tax schedules before and after the reform, as described in Section 3.

Panel A of Figure 8 shows the average, minimum and maximum mechanical change in the MTR, in percentage points, by bins of average reported revenue in 2016–2019. Among small firms, with revenue below T_1, the minimum change is a decrease of 5pp across the whole range, and a maximum increase of 10pp. The average change rises from -5pp to around zero just below T_1. Among medium firms, with revenue between T_1 and T_2, the MTR fell by up to 15pp or did not change, with an average decrease of around 10pp. Large firms, with revenue above T_2, saw no change. Panel B shows the same for the ATR. For small firms, the minimum change is again a decrease of 5pp, while the maximum rises smoothly with revenue to an increase of about 7pp just below T_1, and the average stays negative throughout. For medium firms, the ATR fell for all firms, by between about 2pp and 15pp, with an average decrease of around 13pp. For large firms it did not change.

[Figure 8 here]

Table 4 describes the treatment and control groups in our main sample: pooled by the direction of the change in the MTR, as used in Sections 6.1.1 to 6.3, and for each group separately, as used in Section 6.4. It also reports the number of firms and firm-years in each group, and the firm-years used in the estimations. Of the 12,668 firms in the main sample, 58% faced a lower MTR (G1 and G5) and 9% a higher one (G3 and G4), 14% are small

and medium firms whose MTR did not change but whose ATR fell (G2 and G6), and the remaining 19% are large firms (GC). Since groups are defined by pre-reform revenue and profits, treated and control firms differ in size and profit margins by construction, which is why our specifications control for pre-reform employment and the wage bill to revenue ratio, interacted with year.

[Table 4 here]

Actual tax rates after the reform depend on what firms report, so they can differ from the mechanical ones. Figure 9 compares, for treated firms, the mechanical change in their tax rates with the actual change in 2020, computed from their 2020 reports under the new schedule relative to their 2016–2019 average actual rate. Actual changes move closely with mechanical changes, for both the MTR and the ATR. The mechanical changes therefore capture the direction and relative size of the changes firms faced.

[Figure 9 here]

As a first approach, we consider a binary treatment definition, T_i, where our treatment variable takes the value of one, $T_i = 1$, if the calculated mechanical marginal tax rate change of a firm is different from zero, $\Delta MTR \neq 0$, and zero otherwise. We use this to compare firms that experienced a change in their MTR relative to the control group of large firms (GC). Small and medium firms whose MTR did not change but whose ATR fell (G2 and G6) are excluded from these estimations, since their ATR changed, so they cannot serve as controls.

As a second strategy, we employ a continuous measure based on the net-of-tax rate. For observations before the reform, the value of this variable is equal to the log of one minus the marginal tax rate, $\log(1 - MTR)$, under the previous tax schedule based on the average reported information from 2016–2019. From 2020 onwards, it takes the value of the log of one minus the calculated new MTR after the reform, according again to the average tax reports from 2016–2019. We use this measure to study the sensitivity of reported profits to changes in the net-of-tax rate, and we also use it to explore its effects on reported revenue and costs to disentangle the profit responses.

Finally, we also integrate the ATR into our analysis. We perform a binary DiD to investigate how firms' reported profits responded to different combinations of changes in MTR and ATR for our six treatment groups shown in Figure 7. We also compute elasticity estimates for the different groups.

5.2 Estimation

Reported profits (π_{it}), the tax base and our main outcome of interest, are defined as the difference between reported revenue (y_{it}) and costs (c_{it}). Since firms report either positive or zero profits, we use Poisson pseudo-maximum likelihood (PPML) models throughout our analysis to account for the presence of zeros, as recommended by Silva and Tenreyro (2006) and Chen and Roth (2024). While most of our study focuses on the effects on reported profits, we also investigate the impacts on reported revenue and costs to better understand the anatomy of the observed responses.

We begin by employing a standard binary DiD design. Using a binary treatment allows us to causally identify average responses under a standard parallel trends assumption, without requiring the stronger version of the assumption required for a continuous treatment (Callaway et al., 2024). Our initial model is specified as follows:

$$Z_{it} = \exp\left(\alpha_i + \gamma_t + \beta\, T_i \times \text{Post}_t + X_i'\phi\right) \upsilon_{it} \tag{1}$$

Here, Z_{it} (reported profits, revenue or costs) denotes the outcome of interest for firm i in year t. α_i and γ_t represent firm and year fixed effects, respectively. The vector X_i' includes year fixed effects interacted with pre-reform industry, with deciles of pre-reform average employment, a proxy for firm size, and with deciles of the pre-reform average wage bill to reported revenue ratio, a proxy for firm labour intensity. T_i is a dummy variable indicating whether firm i was treated with the introduction of the reform, according to our treatment definition described in the subsection above, and Post_t is a dummy equal to 1 for the years t following the reform. The year fixed effects help us control for macro-level shocks that could have affected firms' real and reporting behaviour, such as the COVID-19 pandemic. The interaction between year and pre-reform industry fixed effects further accounts for the fact that different industries may have been affected differently by such shocks. Standard errors are clustered at the firm level. Our first analyses capture responses to MTR, as reflected in the notation below, and we use similar specifications to study responses to the ATR.

Our second set of specifications considers the fact that the reform may have led to differential responses by whether tax rates decreased or increased. Hence, we perform separate binary DiD estimations for the group treated with a decrease in their MTR, $T_{\downarrow MTR,i}$ and the group experiencing an increase $T_{\uparrow MTR,i}$, using the same control group of large firms in both estimations:

$$\begin{aligned} Z_{it} &= \exp\left(\alpha_i + \gamma_t + \beta\, T_{\downarrow MTR,i} \times \text{Post}_t + X_i'\phi\right) \upsilon_{it}, \\ Z_{it} &= \exp\left(\alpha_i + \gamma_t + \beta\, T_{\uparrow MTR,i} \times \text{Post}_t + X_i'\phi\right) \upsilon_{it} \end{aligned} \tag{2}$$

We also analyse firms' behaviour over time around the introduction of the reform using an event-study specification of the DiD approach and look at the dynamics of the effects:

$$\begin{aligned} Z_{it} &= \exp\left(\alpha_i + \gamma_t + \textstyle\sum_{\kappa=-4}^{-2} \theta_\kappa T_{\downarrow MTR,i}\, \tau_t^\kappa + \sum_{\kappa=0}^{4} \delta_\kappa T_{\downarrow MTR,i}\, \tau_t^\kappa + X_i'\phi\right)\upsilon_{it}, \\ Z_{it} &= \exp\left(\alpha_i + \gamma_t + \textstyle\sum_{\kappa=-4}^{-2} \theta_\kappa T_{\uparrow MTR,i}\, \tau_t^\kappa + \sum_{\kappa=0}^{4} \delta_\kappa T_{\uparrow MTR,i}\, \tau_t^\kappa + X_i'\phi\right)\upsilon_{it} \end{aligned} \tag{3}$$

Here the coefficients $\{\theta_\kappa\}_{\kappa=-4}^{-2}$ are the event-study DiD pre-trends and $\{\delta_\kappa\}_{\kappa=0}^{4}$ are the treatment effects relative to 2019 (corresponding to the omitted period -1), with $\tau_t^\kappa = \mathbf{1}[t = 2020 + \kappa]$.

Finally, we assess the effects of varying treatment intensities on firms' behaviour and derive elasticity estimates using the continuous net-of-tax rate measure. To do this, we use the following specification:

$$Z_{it} = \exp\left(\alpha_i + \gamma_t + \varepsilon_{1-MTR} \cdot \log\left(1 - MTR_{it}\right) + X_i'\phi\right)\upsilon_{it} \tag{4}$$

6 Results

Our results follow the order set out in Section 4. We first show that reported profits responded to the changes in the MTR in the expected direction, and with similar magnitudes for tax cuts and increases (Section 6.1), and assess identification, robustness and the margins of adjustment (Sections 6.2 and 6.3). We then turn to the question whether firms responded to the MTR or to the ATR (Section 6.4).

6.1 Responses to Marginal Tax Rates

6.1.1 DiD Estimations

The coefficients in line 1 of Panel A in Table 5 present the estimated average effect of being treated by a change in MTR on reported profits, relative to the control group, as derived from the binary DiD analysis. Here, the treatment group consists of small and medium firms that experienced a change in their MTR, while our control group comprises large firms (GC), whose marginal and average tax rates did not change.

[Table 5 here]

For comparability, Columns 1–3 of Panel A present results for the unbalanced panel. This dataset includes all firms observed at least once before and once after the reform, for which

we can construct the treatment[6]. Some of them enter or exit the sample during the period. Columns 4–6 display results for our preferred balanced-panel sample, but without controls. Finally, Columns 7–9 report results for our preferred specification, both for this table and for other upcoming results, which uses the balanced-panel sample and includes controls.

Column 7 of Panel A shows that firms affected by a change in MTR (whether an increase or decrease) increased their reported profits post-reform compared to pre-reform levels, reflecting that most treated firms faced a lower MTR. Columns 8 and 9 further distinguish responses based on the direction of the MTR change. Firms facing a lower MTR increased their reported profits by 0.38 log points (corresponding to an average MTR reduction of 8pp), while those subject to a higher MTR reduced them by 0.33 log points (associated with an average MTR increase of 8pp). These results align with standard after-tax profit maximisation models: reported profits move in the opposite direction to the MTR. Results from the balanced panel without controls (Columns 4 to 6) exhibit similar patterns, with larger coefficients for firms facing a lower and a higher MTR. Controlling for firm size and labour intensity therefore attenuates the estimated effects in both directions. Looking at Columns 1 to 3, the results using the unbalanced panel sample are qualitatively similar to those of our preferred specifications but exhibit larger magnitudes in both directions, particularly for firms facing a lower MTR. This difference may arise because the estimates also now include firms that either entered or exited the sample during the analysis period, and include no controls.

The binary treatment considers only whether firms experienced a change in their marginal tax rates, without accounting for the magnitude of the change. As such, we cannot draw strong conclusions about firms' sensitivity to the size of tax changes. We therefore present elasticity estimates in Subsection 6.1.3. Nonetheless, since the average changes in the MTR are of similar size in both directions, the coefficients of 0.38 and -0.33 log points already point to responses of similar magnitude to tax cuts and tax increases.

6.1.2 Dynamic Effects: DiD Event-Study Estimations

Figure 10 illustrates the dynamics of the findings using the event-study specification of the DiD approach. Panel A shows that firms experiencing a decrease in MTR followed parallel trends with the control group prior to the policy change. Following the reform's introduction, there is a sharp and immediate increase in reported profits, which persists over time. Similarly, Panel B shows that firms subject to an increased MTR also exhibited parallel trends in reported profits before the reform and experienced a decline in reported profits in the first year of the new schedule (2020), which deepened in 2021 and remained stable

[6]See Table 1, Column 1, for descriptive statistics of this sample.

afterwards. Firms thus adjusted in the first year under the new schedule, even though the reform changed several features of the schedule at once.

[Figure 10 here]

Figure A1 presents results for the unbalanced sample without controls. These results are qualitatively consistent with those from our preferred dynamic specifications but exhibit stronger effects. The biggest difference arises for firms experiencing an increase in MTR, where the decline in reported profits intensifies over time. As above, these estimates also include firms that enter or exit the sample.

6.1.3 Elasticity Estimates

Columns 7–9 of Panel B in Table 5 present the reported profit elasticity estimates with respect to changes in the marginal net-of-tax rate, providing a measure of firms' sensitivity that accounts for the size of the tax change. Using the preferred sample, the estimated elasticity is 3.13 for firms whose MTR fell and 3.18 for firms whose MTR rose (Columns 8 and 9): a 1% increase in the net-of-tax rate raises reported profits by about 3.1% to 3.2%. Although these elasticities are large compared to estimates from high-income countries, where values are typically below 1 (Devereux et al., 2014), our findings are consistent with the hypothesis and empirical evidence that elasticities tend to be much higher in developing economies with weaker enforcement capacity (Alejos, 2018; Lobel et al., 2024). Moreover, our results are in line with previous local estimates from Costa Rica under the old tax schedule, derived using bunching techniques, which estimate the elasticities between 2.7 and 4.8 (Bachas and Soto, 2021).

The two elasticities are very similar, and the difference is small relative to the precision of each estimate. This contrasts with findings in the literature on other types of taxes, which show that agents respond asymmetrically to tax increases and decreases (Benzarti et al., 2020; Saez et al., 2019, 2021; Hussain and Malik, 2016; Wu and Yang, 2024). In our setting, the response of reported profits does not depend on the direction of the tax change, so the same elasticity can be used to assess tax cuts and tax increases.

The design of the reform allows this study, to our knowledge, to be the first to compare responses to tax increases and decreases within a single reform event, rather than across tax changes at different points in time, as done in previous research. The similarity of the responses therefore cannot be explained by differences in the timing or circumstances of separate reforms.[7]

[7]The firms facing tax cuts and tax increases are different. The elasticities are, however, also similar for

6.1.4 Heterogeneity Analysis

We also perform heterogeneity analysis by different types of firms. We first examine heterogeneity by pre-reform firm size. To do this, we implement a triple-difference specification by adding an interaction term between the treatment effect and dummies indicating different firm sizes. Specifically, we divide the potentially treated small and medium firms into four groups based on pre-reform revenues. The cut-off points for these groups are $T_1/2$, T_1, $T_1 + (T_2 - T_1)/2$, and T_2, where T_1 and T_2 are the CIT revenue thresholds before the reform. This yields the following groups: *smaller* small, *larger* small, *smaller* medium, and *larger* medium. The base (excluded) category is the first one.

The treatment effect reported in the first line of results in Table A2 reflects the effect for smaller small firms, while the triple-difference coefficients below it indicate how that effect changes relative to this baseline across the other size groups. Looking at our preferred specifications, particularly Column 8, which focuses on firms that saw a decrease in their MTR, the results show that the positive effect is equally large for both small firm categories. For medium firms, the effect remains positive, but its magnitude is smaller. In the case of MTR increases, shown in Column 9 (note that the last two medium firm size categories are excluded here, as only small firms experienced MTR increases; see Figure 8), the negative effect is similar for both small firm categories. Overall, the evidence suggests that small firms experienced the strongest responses to the reform.

Second, we perform a heterogeneity analysis by the eleven industry categories. The base (excluded) category is education, human health & social work. Thus, the treatment effect $T \times \text{Post}$ in the first line of results in Table A3 reflects the effect for the base industry, while the triple-difference coefficients show how that effect varies relative to this category across the other industries. Column 8, which shows the effect of the decrease in MTR, indicates that the base category saw a positive and significant effect. The industry categories of construction; information & communication; financial & insurance; and real estate appear to have a larger positive effect, particularly financial & insurance. The other categories do not show significant differences compared to the base category.

Column 9 shows the effects for firms that saw an increase in their MTR. Here, the effect is negative but not significant for the baseline category education, human health & social work. The decline in reported profits is larger, and significant, in agriculture, forestry & fishing; manufacturing, mining & other industries; wholesale, retail, transportation & storage; professional, scientific, technical & administrative; and other services.

medium firms whose MTR fell and for small firms whose MTR rose, between 3.3 and 3.5 (Section 6.4). The exception is the small firms with the lowest profits, whose elasticity is larger.

6.2 Identification and Robustness

In this section, we test the assumptions behind our specification and the robustness of our results. We first examine the two identifying assumptions of the difference-in-differences design, no anticipation and parallel trends. We then address two related concerns, mean reversion and differential trends between groups. Next, we show that our results do not depend on the sample, by excluding firms that bunched below the revenue thresholds before the reform and by using the unbalanced panel. Finally, we test whether two shocks that coincided with the reform, the extension of VAT to services and the COVID-19 pandemic, drive our estimates.

We first test the no-anticipation assumption by conducting placebo tests, estimating the event-study DiD specification while setting 2018 and 2019 fiscal years as hypothetical treatment start years instead of the 2020 fiscal year. As shown in Figure A2, there is no significant effect in either of the treatment groups for 2018 or 2019, while significant effects emerge from 2020. Although the reform was announced at the end of 2018, firms do not appear to have adjusted their behaviour before its enactment in the 2020 fiscal year. In addition, our treatment variables are constructed using tax reports from 2016 to 2019, covering three years of tax declarations filed before the new tax schedule was known. This helps mitigate concerns about anticipation effects and self-selection into treatment or control groups.

We then assess parallel trends by visually comparing the evolution of profits between the treatment and control groups before the reform. Figure 10 shows that both groups followed similar trajectories prior to the policy change, and we cannot reject that the pre-reform event-study coefficients are jointly zero (p-values of 0.179 and 0.741 for firms facing a lower and a higher MTR). In addition, Figure A3 presents raw event studies, showing that reported profits for each treatment and control groups follow close trends between 2016 and 2019 and diverge after the reform.

Three steps are taken to mitigate concerns about mean reversion and differential trends between groups. First, as explained in Section 5.1, treatment is constructed in all our specifications from four years of pre-reform reports instead of a single year. This reduces the extent to which a firm's assigned treatment group reflects transitory shocks, after which its outcomes would revert to their usual level for reasons unrelated to the reform. Second, as a robustness check, we add controls for log pre-reform revenue interacted with year. This allows firms of different pre-reform size to follow different paths, and also absorbs part of any reversion related to their pre-reform revenue. Third, we also allow treated and control firms to follow different linear trends. Panel B of Table 6 reports the binary DiD estimates under these two checks. Controlling for pre-reform revenue brings the effects of tax cuts and

tax increases closer together. With group-specific trends, both effects are attenuated by a similar amount, but they still move in the direction implied by the change in the MTR, remain significant and are of similar magnitude to the baseline estimates.

We also exclude firms whose pre-reform revenue lies in the range where firms bunched below the two revenue thresholds to see if these firms that were potentially adjusting their reported revenue in response to the notches before the reform affect our estimates. Moreover, we also run our estimations using the unbalanced panel with employment data sample, therefore including firms that exited or entered through our analysis period. In both cases (Table 6, Panels B and C), the results for tax cuts and increases remain qualitatively similar compared to the baseline.

The same law that introduced the CIT reform replaced the sales tax with a value-added tax that also covers services, which brought new firms into the consumption tax net. We therefore run a triple-difference specification, interacting the treatment with an indicator for firms in industries that became liable for VAT with the reform, to explore if this affected the outcome. The interaction is not significant for either group, and the effects remain of similar size (Table 6, Panel D).

Finally, we explore whether our estimates are affected by the COVID-19 shock, which overlapped with the CIT reform introduced in the fiscal year 2020 (starting in October 2019), and the sanitary measures that restricted mobility and in-person activities from March 11, 2020, in Costa Rica. As discussed earlier, our preferred specifications include year and industry-year fixed effects to account for time specific economic trends and sector-specific impacts of the pandemic. Here, we provide further tests to show that our results are driven by the reform rather than the pandemic. First, we run a triple-difference specification by adding an interaction term between the treatment effect and a dummy indicating whether a firm belongs to one of the three industry groups most affected by the COVID-19 shock. We define these industries as those with a GDP activity drop of more than 10% between 2019 and 2020: accommodation & food services (-36.3%), other services (-18.1%), and wholesale, retail, transportation & storage (-10.8%)[8] Panel D in Table 6 shows that the interaction is not significant for firms facing a higher MTR, though for firms facing a lower MTR, the effect is somewhat smaller in the most affected industries (-0.12), but remains large and positive. Second, as an additional check, we exclude the two years most affected by the pandemic, 2020 and 2021 fiscal years, from our estimations. Figure A5 shows that the results remain consistent with our preferred estimates, which is also confirmed in Panel D of Table 6 (0.40 and -0.36).

[8]See Table A4 for 2020 GDP growth data by industry.

[Table 6 here]

6.3 Anatomy of the Response

To disentangle the profit response, we examine how firms adjust their revenue and costs in reaction to marginal tax changes. Table 7 presents the elasticity estimates of reported profits (the same as our preferred estimates in Table 5, for comparability) alongside the elasticity estimates for reported revenue and costs, all with respect to the net-of-tax rate.

[Table 7 here]

For firms that experienced a decrease in their MTR, the estimated elasticity is 3.13, driven by positive responses in both reported revenue and costs. However, the response in reported revenue was larger in magnitude (elasticity of 0.40) than the response in reported costs (elasticity of 0.26), which accounts for the overall net increase in reported profits.

For firms that experienced an increase in the MTR, the estimated reported profit elasticity is 3.18, indicating that a higher tax rate (equivalently, a lower marginal net-of-tax rate) leads to a decline in reported profits. In this case, both reported revenue and cost elasticity estimates are negative, at -0.91 and -2.37, respectively. A higher MTR thus led to higher reported revenue and costs, but costs rose by more, so reported profits fell. One likely explanation is that this treated group consists of small firms with relatively high profits (corresponding to Groups G3 and G4 in Figure 7), located near the notch at the lower revenue threshold, T_1. The replacement of the notch at T_1 may have unlocked growth opportunities. These firms expanded their revenue reports, but the tax rate increase triggered an even stronger response in cost adjustments, ultimately leading to lower reported profits.

The larger magnitude of the reported costs response for firms with an increased MTR is also consistent with the findings of Bachas and Soto (2021), who study how Costa Rican firms respond to higher tax rates in a static setting. They find that firms adjust more on the reported costs side than on the revenue side when facing higher tax rates, providing evidence that these adjustments largely stem from tax evasion and not from real economic responses. The argument is that costs are easier to manipulate than revenue, making them a preferred channel for firms seeking to minimise tax liabilities. The fact that the reform led to reported revenue growth in both groups of firms also suggests that replacing the notch at T_1 removed distortions that had previously held back firms' expansion. Hence, while the response of reported profits is of similar magnitude for tax cuts and tax increases, firms adjust through

different margins: mainly revenue when the MTR falls, and mainly costs when it rises.

6.4 Responses to Marginal Versus Average Tax Rates

In this section, we turn to the question of whether changes in the ATR, alongside those in the MTR, influenced firms' reported profit responses. We compare each group of small and medium firms defined in Figure 7 (G1 to G6) with the control group of large firms (GC), whose tax rates did not change. As discussed in Section 4, ironing and a response to the MTR lead to different predictions for G3, G2 and G6 (Table 3).

Panel A of Table 8 presents the binary DiD analysis for the six treated groups, while the dynamics are illustrated in Figure 11, obtained via the event-study DiD specifications. Both Group G1 (small firms) and Group G5 (medium firms) experienced decreases in both their MTR and ATR, leading, as expected, to an increase in reported profits following the reform, as shown in Columns 1 and 5. The magnitude of this effect appears slightly larger for G1, consisting of small firms with the lowest profits. In contrast, Group G4 shown in Column 4, which faced increases in both rates, saw a decline in reported profits, as anticipated. Group G3 experienced an increase in MTR but a reduction in ATR, so ironing would have led these firms to increase their reported profits. However, as indicated in Column 3, G3 followed the theoretical prediction by reducing reported profits, although by less than G4.

[Table 8 here]

[Figure 11 here]

Next, we examine Groups G2 (small firms) and G6 (medium firms). In both groups the reform left the MTR unchanged but reduced the ATR. Theory therefore predicts no change in their reported profits. The results for G2 confirm this prediction: the coefficient on being treated with only a decrease in the ATR (and no change in the MTR) is small and insignificant. G6, however, shows a significant negative effect, even though its MTR did not change. Although this deviates from the theoretical prediction, it cannot reflect a response to the lower ATR, since that response would have increased reported profits, not decreased them.

A possible explanation for the decline in G6 could be a response to the revenue notch the reform kept in place, T_2. As the reform lowered the ATR of G6 firms while leaving the rate above T_2 unchanged, the jump in tax liability from crossing T_2 became larger for them, potentially strengthening the incentive to stay below it. If this drove our result, the

decline should be concentrated among the G6 firms closest to T_2. However, we find this is not the case: when we progressively exclude the firms with the highest pre-reform revenue, the negative effect does not shrink, as shown in Figure A6.

Panel B of Table 8 presents the reported profit elasticity estimates in response to changes in the marginal net-of-tax rate for the groups that experienced a change in their MTR (G1, G3, G4, and G5). Given our treatment definition, groups G2 and G6 did not experience a change in their mechanical marginal net-of-tax rate; therefore, we cannot compute these estimates for them. These results reflect the symmetry discussed in Subsection 6.1.3. Firms that faced a decrease in their MTR (G1 and G5) and firms that faced an increase (G3 and G4) respond with elasticities of similar magnitude: 3.46 for G5, and 3.32 and 3.37 for G3 and G4, respectively, in line with the pooled estimates of 3.13 for decreases and 3.18 for increases. The exception is G1, the smallest firms facing a decrease in their MTR, whose elasticity is larger, at 8.62.

These findings provide no evidence that the reform led firms in Costa Rica to deviate from after-tax profit-maximising behaviour, despite operating under a CIT schedule in which the marginal and average tax rates no longer coincide, and despite multiple simultaneous changes in tax design. Reported profits respond to the MTR, not to the ATR, and they do so symmetrically to rate cuts and rate increases. This result contrasts with previous empirical evidence showing that individuals tend to iron non-linear schedules, responding to average tax rates or prices, even when models of optimal behaviour predict responses to marginal ones (de Bartolomé, 1995; Ito, 2014; Rees-Jones and Taubinsky, 2020; Shaffer, 2020; Ho, 2023), and that they are often confused by or inattentive to the structure of tax schedules (Chetty et al., 2009; Feldman et al., 2016; Abeler and Jäger, 2015).

We propose two potential, non-exclusive explanations for these findings. First, firms are generally considered more sophisticated economic agents than individuals and may therefore face fewer challenges or costs when responding to the incentives of less simple schedules. For example, they are more likely to rely on tax preparers and accountants, who reduce the costs of complexity and the risk of misinterpretations (Zwick, 2021), and through whom knowledge about tax incentives spreads (Chetty et al., 2013). Second, the details of the reform were widely publicised, and information was readily available on the website of the *Ministerio de Hacienda* (Ministry of Finance), where firms file their tax returns. This may have reduced information frictions, which are known to dampen responses to tax incentives (Chetty et al., 2013; Kostl and Myhre, 2021). Public interest in the reform was also evident, as suggested by a surge in Google searches for the term *Ministerio de Hacienda* around the announcement and implementation period (see Figure A7). This suggests that well-communicated tax reforms, such as Costa Rica's CIT reform, can help tax authorities implement less distor-

tionary systems while minimising errors and adjustment costs, even when the new schedule is less simple.

7 Conclusion

This paper examines how firms respond to tax incentives when a simple but distortionary tax schedule is replaced by a less simple one in a developing economy. We study Costa Rica's 2020 corporate income tax reform, which, for small and medium firms, replaced a notched schedule based on revenue with a kinked schedule based on profits. Using administrative tax returns from 2016 to 2024 and a difference-in-differences design that exploits the different changes in marginal and average tax rates across firms, we find that firms responded quickly to the reform and in line with after-tax profit maximisation.

Reported profits rose where the MTR fell and fell where it rose, with similar elasticities of 3.13 and 3.18. Firms adjusted mainly through revenue when the MTR fell, and mainly through costs when it rose. Importantly, firms reacted to the right incentive: reported profits followed changes in the MTR, not in the ATR, so we find no evidence of the average-rate mistakes documented for individuals. At the same time, bunching and cost inflation disappeared at the replaced notch, persisted at the remaining one, and little or no bunching emerged at the new profit thresholds.

These findings suggest that the trade-off between simplicity and efficiency in tax design may be less severe than the evidence from individuals implies. Firms in a developing country, facing several changes in their schedule at once and several rates applying to different portions of their profits, identified the relevant marginal incentive, and a less simple schedule removed a major distortion without creating new ones. The symmetry of responses also suggests that, unlike the evidence for other taxes, a single elasticity can guide corporate tax changes in either direction.

Future work may extend these results to the extensive margin, which we expect to be secondary in this setting since firms can adjust what they report, and to the revenue consequences of the reform. Taken together, our evidence suggests that less simple, and more economically efficient, tax designs should be considered in developing countries.

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Tables

Table 1. Samples and Pre-Reform Characteristics

	Unbalanced panel		Balanced panel	
	All firms	Employment records	All firms	Employment records
	(1)	(2)	(3)	(4)
2016–2019 averages				
Revenue (million CRC)	48.6	59.7	59.9	68.1
	(43.4)	(46.1)	(46.1)	(47.3)
Costs (million CRC)	45.7	57.0	55.8	64.1
	(43.8)	(46.7)	(46.1)	(47.5)
Reported profits (million CRC)	5.2	5.8	6.2	6.6
	(7.2)	(7.6)	(7.3)	(7.5)
Profit margin	0.14	0.12	0.13	0.12
	(0.17)	(0.15)	(0.14)	(0.13)
Years with zero profits (%)	19.3	21.1	15.0	16.3
	(31.9)	(32.5)	(26.4)	(27.3)
Employment (employees)		2.6		2.7
		(2.7)		(2.7)
Wage bill (million CRC)		10.1		11.1
		(12.7)		(12.8)
Share of firms (%)				
With employment records	58.8	100.0	70.6	100.0
Small	67.8	57.3	57.5	49.4
Medium	21.5	27.6	27.1	31.4
Large	10.7	15.2	15.4	19.2
Firms	49,827	29,302	17,952	12,668
Firm-years	348,340	215,075	161,568	114,012

Notes: This table compares the samples used in the paper. All samples include firms registered as active, with positive reported revenue and costs, observed at least once before and once after the reform, with reported revenue above 3 million CRC in at least one pre-reform year and maximum pre-reform revenue below 200 million CRC. The balanced panel keeps firms observed in every year from 2016 to 2024. Employment records refers to firms with formal employees reported to the Social Security Authority in at least one year between 2016 and 2019. Column 4 is our main sample; Column 1 is used in Columns 1–3 of Table 5, and Column 2 in the robustness checks of Table 6. The table reports means of each firm's 2016–2019 averages, with standard deviations across firms in parentheses; employment and wage bill are shown only for samples restricted to firms with employment records. Values are in million 2014 CRC and winsorised at the 1st and 99th percentiles within each year. The profit margin is the ratio of reported profits to reported revenue. Reported profits are zero for firms that report losses, so average costs can exceed average revenue. Small, medium and large firms have pre-reform average revenue below the first revenue threshold of the pre-reform schedule, between the two thresholds, and above the second threshold (Section 3).

Table 2. Summary CIT Before and After the Reform

	Small and Medium Firms		Large Firms	
	Before	After	Before	After
Tax Base	Profits	Profits	Profits	Profits
Tax Rate function of	Revenue	Profits	Revenue	Revenue
Design	Notch	Kink	Notch	Notch
Tax Rates (%)	10, 20 (ATR)	5, 10, 15, 20 (MTR)	30 (ATR)	30 (ATR)

Notes: This table summarises the key features of Costa Rica's CIT system before and after the 2020 reform. The pre-reform system applied a notched design, where the average tax rate on reported profits was determined by reported revenue thresholds. Small and medium firms faced lower average tax rates of 10% and 20%, while large firms faced the statutory 30% average tax rate. After the reform, small and medium firms moved to a kinked schedule where the marginal tax rate applies to profits above certain thresholds, with rates of 5%, 10%, 15%, and 20%. Large firms continued under the same 30% average tax rate, with no changes in their tax schedule.

Table 3. Predicted Effects on Reported Profits, by Group

	$\downarrow MTR$		$\uparrow MTR$		$\leftrightarrow MTR$		Control
	G1 Small	G5 Medium	G3 Small	G4 Small	G2 Small	G6 Medium	GC Large
Change in MTR	$\downarrow$	$\downarrow$	$\uparrow$	$\uparrow$	$\leftrightarrow$	$\leftrightarrow$	$\leftrightarrow$
Change in ATR	$\downarrow$	$\downarrow$	$\downarrow$	$\uparrow$	$\downarrow$	$\downarrow$	$\leftrightarrow$
Predicted effect on profits	$\uparrow$	$\uparrow$	$\downarrow$	$\downarrow$	$\leftrightarrow$	$\leftrightarrow$	$\leftrightarrow$

Notes: This table summarises the direction of the changes in the marginal and average tax rates of each group for an unchanged tax report (Figure 7) and the effect on reported profits predicted if firms respond to the marginal rate, as after-tax profit maximisation implies. Average changes in each rate, in percentage points, are reported in Table 4. $\downarrow$ decrease, $\uparrow$ increase, $\leftrightarrow$ no change.

Table 4. Treatment and Control Groups

	$\Delta MTR \neq 0$			Treatment groups						Control
	All	$\downarrow MTR$	$\uparrow MTR$	G1	G2	G3	G4	G5	G6	GC
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
2016–2019 averages										
Revenue (million CRC)	47.0	48.3	38.9	26.7	33.5	35.9	41.7	80.5	81.9	146.2
	(28.6)	(30.3)	(10.6)	(14.2)	(11.9)	(11.1)	(9.3)	(16.0)	(15.7)	(23.7)
Costs (million CRC)	45.8	49.2	25.1	28.4	27.8	26.6	23.7	80.2	62.3	139.3
	(32.1)	(33.1)	(10.8)	(21.1)	(12.5)	(11.4)	(10.1)	(21.6)	(19.0)	(27.2)
Reported profits (million CRC)	4.4	2.8	14.0	2.0	6.4	9.7	17.9	4.0	19.4	9.1
	(5.0)	(2.4)	(5.9)	(1.5)	(0.8)	(1.1)	(5.8)	(2.8)	(8.1)	(9.3)
Profit margin	0.12	0.08	0.36	0.09	0.22	0.29	0.43	0.05	0.24	0.06
	(0.13)	(0.07)	(0.16)	(0.09)	(0.10)	(0.12)	(0.16)	(0.04)	(0.11)	(0.07)
Years with zero profits (%)	19.5	22.0	4.2	25.4	7.7	4.5	4.0	16.9	4.3	12.6
	(29.5)	(30.7)	(12.5)	(32.1)	(15.8)	(12.3)	(12.8)	(27.7)	(12.0)	(24.1)
Employment (employees)	2.2	2.4	1.5	1.7	1.6	1.6	1.4	3.4	2.6	4.9
	(2.2)	(2.3)	(1.3)	(1.6)	(1.4)	(1.5)	(1.0)	(2.7)	(2.1)	(3.7)
Wage bill (million CRC)	8.4	8.9	5.5	6.0	5.9	5.7	5.3	13.3	11.5	21.6
	(9.9)	(10.4)	(5.3)	(7.3)	(5.8)	(6.0)	(4.7)	(12.5)	(10.3)	(17.7)
Mechanical change in tax rates (pp)										
ΔMTR	-5.8	-8.1	8.3	-5.0	0.0	6.3	10.0	-12.8	0.0	0.0
ΔATR	-7.5	-8.8	0.7	-5.0	-4.1	-1.8	3.0	-14.5	-6.9	0.0
Firms	8,470	7,291	1,179	4,364	721	556	623	2,927	1,046	2,431
Firm-years	76,230	65,619	10,611	39,276	6,489	5,004	5,607	26,343	9,414	21,879
Firm-years in estimations	74,250	63,639	10,611	37,827	6,489	5,004	5,607	25,812	9,414	21,681

Notes: This table describes the treatment and control groups in our main sample, the balanced panel of firms with employment records. Groups are defined by the mechanical changes in the marginal and average tax rates, calculated from 2016–2019 average revenue and reported profits (Figure 7). Column 1 includes all firms whose marginal tax rate changed, Column 2 those whose marginal tax rate fell (G1 and G5) and Column 3 those whose marginal tax rate rose (G3 and G4). The table reports means of each firm's 2016–2019 averages, with standard deviations across firms in parentheses, and the average mechanical change in each tax rate in percentage points. Values are in million 2014 CRC and winsorised at the 1st and 99th percentiles within each year. Employment is the average number of formal employees per month reported to the Social Security Authority. Reported profits are zero for firms that report losses, so average costs can exceed average revenue. Firm-years in estimations excludes firms that report zero profits in every year, which the Poisson model drops; each estimation also includes the firm-years of the control group.

Table 5. Marginal Tax Rate Effects on Reported Profits

	(1) $\downarrow MTR+\uparrow MTR$	(2) $\downarrow MTR$	(3) $\uparrow MTR$	(4) $\downarrow MTR+\uparrow MTR$	(5) $\downarrow MTR$	(6) $\uparrow MTR$	(7) $\downarrow MTR+\uparrow MTR$	(8) $\downarrow MTR$	(9) $\uparrow MTR$
PANEL A: DiD									
$T \times$ Post	0.1550***	0.5102***	-0.4221***	0.1225***	0.4211***	-0.4377***	0.2103***	0.3802***	-0.3268***
	(0.0181)	(0.0184)	(0.0211)	(0.0251)	(0.0257)	(0.0321)	(0.0307)	(0.0309)	(0.0485)
Implied ITT	0.1676	0.6656	-0.3444	0.1303	0.5236	-0.3545	0.2341	0.4625	-0.2788
	(0.0211)	(0.0306)	(0.0138)	(0.0283)	(0.0392)	(0.0207)	(0.0379)	(0.0452)	(0.0350)
Treated baseline average	3.79	2.20	14.60	4.37	2.80	14.05	4.37	2.80	14.05
Δ MTR (pp)	-5.10	-7.10	8.35	-5.83	-8.11	8.26	-5.83	-8.11	8.26
Balanced panel	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes
Firms	42,972	38,014	10,189	10,659	9,480	3,588	10,659	9,480	3,588
N	300,833	266,019	74,558	95,931	85,320	32,292	95,931	85,320	32,292
PANEL B: Elasticity									
ε_{1-MTR}	4.4023***	3.6661***	4.0595***	4.0170***	3.6230***	4.1904***	3.7719***	3.1311***	3.1824***
	(0.0775)	(0.1442)	(0.1927)	(0.1290)	(0.2011)	(0.3025)	(0.1334)	(0.2048)	(0.4244)
Treated baseline average	3.79	2.20	14.60	4.37	2.80	14.05	4.37	2.80	14.05
Δ MTR (pp)	-5.10	-7.10	8.35	-5.83	-8.11	8.26	-5.83	-8.11	8.26
Balanced panel	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes
Firms	42,972	38,014	10,189	10,659	9,480	3,588	10,659	9,480	3,588
N	300,833	266,019	74,558	95,931	85,320	32,292	95,931	85,320	32,292

Notes: This table presents difference-in-differences estimates of the effects of changes in the marginal tax rate on reported profits for all treated firms, firms that experienced a decrease in the marginal tax rate and firms that experienced an increase, relative to the control group of large firms (GC). Treatment is determined using the mechanical marginal tax rate change, calculated from 2016–2019 average revenue and reported profits; Groups G2 and G6 are excluded. Columns 1–3 use the unbalanced panel of all firms and Columns 4–9 the balanced panel of firms with employment records. A Poisson pseudo-maximum likelihood model is employed to account for zeros in the dependent variable. All estimations incorporate firm and year fixed effects; additional controls are year fixed effects interacted with pre-reform industry, employment deciles and wage bill-to-revenue ratio deciles. Panel A reports the coefficient on $T \times$ Post, β, and the implied intention-to-treat effect, $\exp(\beta) - 1$; Panel B reports the elasticity with respect to the net-of-tax rate, $\log(1 - MTR)$. Firms and N include the control group. Standard errors (in parentheses) are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6. Robustness of the Marginal Tax Rate Effects on Reported Profits

	(1) $\downarrow MTR + \uparrow MTR$	(2) $\downarrow MTR$	(3) $\uparrow MTR$
PANEL A: Baseline			
Baseline	0.210***	0.380***	−0.327***
	(0.031)	(0.031)	(0.048)
N	95,931	85,320	32,292
PANEL B: Identification			
+ group-specific linear trends	0.190***	0.310***	−0.253***
	(0.045)	(0.048)	(0.072)
+ log pre-reform revenue	0.178***	0.313***	−0.333***
	(0.036)	(0.036)	(0.094)
N	95,931	85,320	32,292
Excluding bunchers	0.223***	0.373***	−0.387***
	(0.033)	(0.032)	(0.055)
N	87,219	78,588	29,331
PANEL C: Sample			
Unbalanced, employment records	0.232***	0.427***	−0.368***
	(0.025)	(0.026)	(0.041)
N	182,740	163,707	52,735
PANEL D: Confounders			
Excluding 2020–21	0.224***	0.402***	−0.355***
	(0.035)	(0.035)	(0.054)
N	74,347	66,094	25,060
$T \times$ Post	0.216***	0.427***	−0.303***
	(0.037)	(0.037)	(0.054)
$T \times$ Post$\times$ most affected by COVID-19	−0.016	−0.118**	−0.081
	(0.051)	(0.052)	(0.073)
N	95,931	85,320	32,292
$T \times$ Post	0.260***	0.340***	−0.391***
	(0.041)	(0.041)	(0.073)
$T \times$ Post$\times$ newly liable for VAT	−0.082	0.068	0.094
	(0.050)	(0.051)	(0.077)
N	95,256	84,690	32,148

Notes: This table presents robustness checks for the difference-in-differences estimates of the effects of changes in the marginal tax rate on reported profits, for all treated firms, firms that experienced a decrease in the marginal tax rate and firms that experienced an increase, relative to the control group of large firms (GC). Treatment is determined using the mechanical marginal tax rate change, calculated from 2016–2019 average revenue and reported profits; Groups G2 and G6 are excluded. A Poisson pseudo-maximum likelihood model is employed to account for zeros in the dependent variable. All estimations incorporate firm and year fixed effects, and year fixed effects interacted with pre-reform industry, employment deciles and wage bill-to-revenue ratio deciles. Panel A reports the preferred specification, which uses the balanced panel of firms with employment records. Panel B adds group-specific linear trends or log pre-reform revenue interacted with year, and excludes firms with pre-reform revenue up to 5 million CRC below the first threshold and 4 million below the second. Panel C uses the unbalanced panel of firms with employment records (Table 1, Column 2). Panel D excludes 2020 and 2021, and adds triple-difference terms with indicators for the industries most affected by COVID-19 (Table A4) and for firms that became liable for VAT with the reform. N includes the control group. Standard errors (in parentheses) are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 7. Anatomy of the Response: Profits, Revenue and Costs

	(1) $\downarrow MTR+\uparrow MTR$ on Profits	(2) $\downarrow MTR$ on Profits	(3) $\uparrow MTR$ on Profits	(4) $\downarrow MTR+\uparrow MTR$ on Revenue	(5) $\downarrow MTR$ on Revenue	(6) $\uparrow MTR$ on Revenue	(7) $\downarrow MTR+\uparrow MTR$ on Costs	(8) $\downarrow MTR$ on Costs	(9) $\uparrow MTR$ on Costs
ε_{1-MTR}	3.7719***	3.1311***	3.1824***	0.3921***	0.3972***	−0.9056***	0.0463	0.2570**	−2.3684***
	(0.1334)	(0.2048)	(0.4244)	(0.0942)	(0.1199)	(0.3239)	(0.1004)	(0.1196)	(0.3736)
Balanced panel	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	95,931	85,320	32,292	98,109	87,498	32,490	98,109	87,498	32,490

Notes: This table presents elasticity estimates of reported profits, revenue and costs with respect to the net-of-tax rate, $\log(1 - MTR)$, for all treated firms, firms that experienced a decrease in the marginal tax rate and firms that experienced an increase, relative to the control group of large firms (GC), using our preferred balanced-panel sample. Treatment is determined using the mechanical marginal tax rate change, calculated from 2016–2019 average revenue and reported profits; Groups G2 and G6 are excluded. A Poisson pseudo-maximum likelihood model is employed to account for zeros in the dependent variable. All estimations incorporate firm and year fixed effects, and year fixed effects interacted with pre-reform industry, employment deciles and wage bill-to-revenue ratio deciles. Samples are smaller for profits because firms reporting zero profits in every year are dropped. N includes the control group. Standard errors (in parentheses) are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

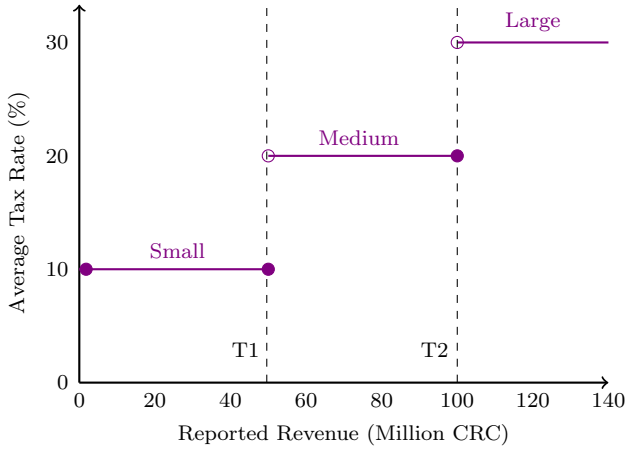
Table 8. Marginal and Average Tax Rate Effects on Reported Profits

	(1) Group G1	(2) Group G2	(3) Group G3	(4) Group G4	(5) Group G5	(6) Group G6
Change in MTR	↓	↔	↑	↑	↓	↔
Change in ATR	↓	↓	↓	↑	↓	↓
PANEL A: DiD						
$T \times$ Post	0.4659***	−0.0957	−0.2590***	−0.3967***	0.3600***	−0.2808***
	(0.0483)	(0.0590)	(0.0606)	(0.0553)	(0.0325)	(0.0370)
Implied ITT	0.5934	−0.0912	−0.2282	−0.3275	0.4333	−0.2448
	(0.0770)	(0.0536)	(0.0468)	(0.0372)	(0.0467)	(0.0280)
PANEL B: Elasticity						
ε_{1-MTR}	8.6170***	–	3.3236***	3.3682***	3.4648***	–
	(0.8934)		(0.7212)	(0.4697)	(0.2188)	
Δ MTR (pp)	−5.0	0	+6.3	+10.0	−12.8	0
Δ ATR (pp)	−5.0	−4.1	−1.8	+3.0	−14.5	−6.9
Treated baseline average	1.97	6.44	9.67	17.96	4.04	19.51
Balanced panel	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes
Firms	6,612	3,130	2,965	3,032	5,277	3,455
N	59,508	28,170	26,685	27,288	47,493	31,095

Notes: This table presents difference-in-differences estimates of the effects of the reform on reported profits for each of the six treatment groups, defined by the changes in their marginal and average tax rates, relative to the control group of large firms (GC), using our preferred balanced-panel sample. Treatment is determined using the mechanical tax rate changes, calculated from 2016–2019 average revenue and reported profits. A Poisson pseudo-maximum likelihood model is employed to account for zeros in the dependent variable. All estimations incorporate firm and year fixed effects, and year fixed effects interacted with pre-reform industry, employment deciles and wage bill-to-revenue ratio deciles. Panel A reports the coefficient on $T \times$ Post, β, and the implied intention-to-treat effect, $\exp(\beta)-1$. Panel B reports the elasticity with respect to $\log(1-MTR)$, not defined for G2 and G6, whose marginal tax rate did not change. Firms and N include the control group. Standard errors (in parentheses) are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

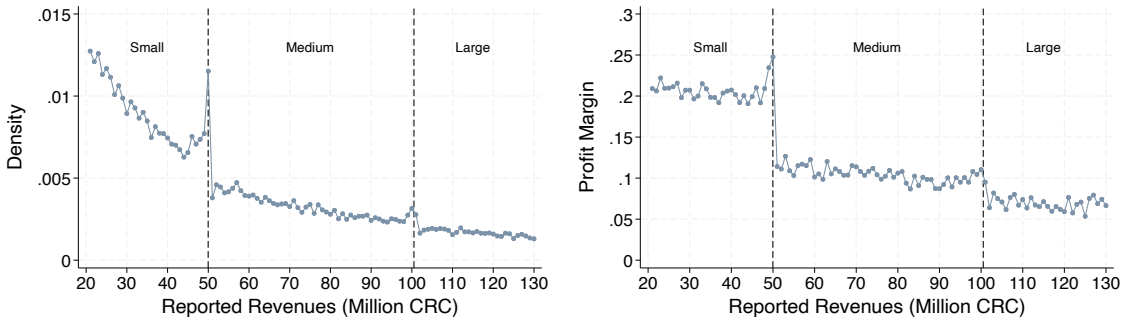
Figures

Figure 1. Corporate Tax Schedule Before the Reform



Notes: This figure shows the notched corporate tax schedule faced by all firms before the 2020 reform. Firms faced increasing average tax rates on their reported profits, which were a function of their revenues. *Small* firms with revenues up to the first threshold T_1 faced an average tax rate of 10 percent, *medium* firms with revenues beyond the first threshold and up to the second threshold T_2 faced a rate of 20 percent, and *large* firms with revenues beyond the second threshold faced an average tax rate of 30 percent.

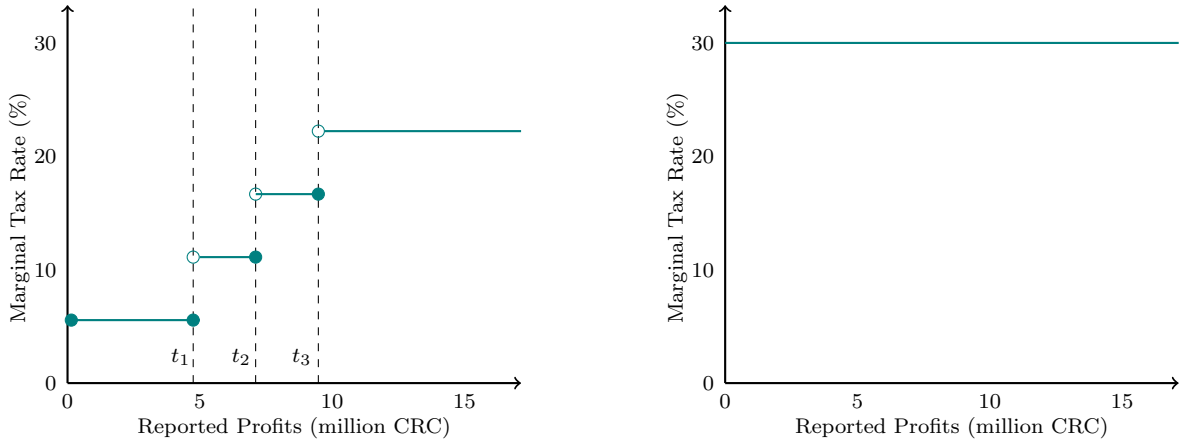
Figure 2. Firm Density and Average Profit Margin Before the Reform



A. Firm density B. Average profit margin

Notes: This figure shows the key patterns of the corporate tax data before the reform, specifically for the years 2016, 2017 and 2018. Panel A displays firm density by revenue bin. Panel B shows average profit margin by revenue bin, where profit margin is defined as profit over revenue. Bin size is 1 million 2014 CRC.

Figure 3. Corporate Tax Schedule After the Reform

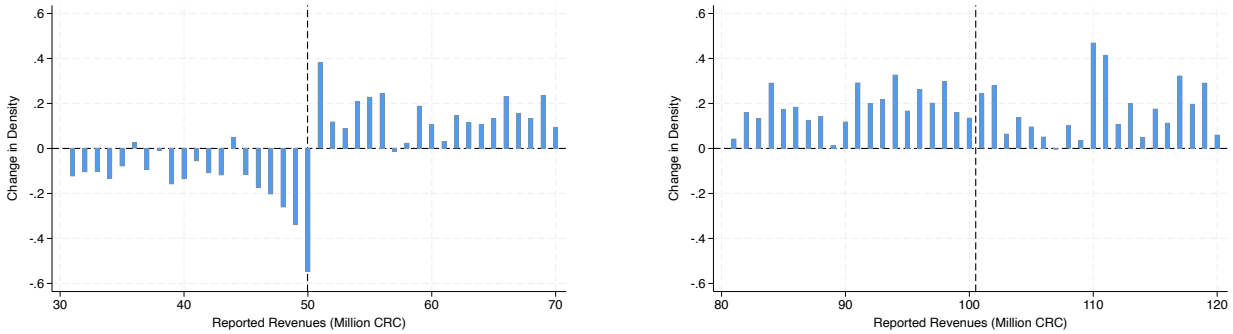


A. Small and medium firms (Revenues $\leqslant T_2$)

B. Large firms (Revenues$> T_2$)

Notes: Panel A shows the kinked profit tax schedule faced by small and medium sized firms after the 2020 reform. Small and medium firms face increasing marginal tax rates on their reported profits, which are a function of their reported profits. Firms with reported profits up to the first threshold t_1 face a marginal tax rate of 5 percent, firms with reported profits beyond the first threshold and up to the second threshold t_2 face a rate of 10 percent, firms with reported profits beyond the second threshold and up to the third threshold t_3 face a rate of 15 percent, and firms with reported profits beyond the third threshold face a marginal tax rate of 20 percent. Panel B shows that large firms kept the same average and marginal tax rate on reported profits of 30 percent, here expressed by level of reported profits.

Figure 5. Firm Density Changes

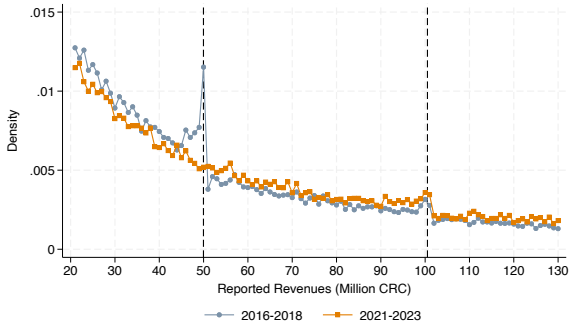


A. Around T_1 threshold

B. Around T_2 threshold

Notes: This figure illustrates changes in accumulated firm density by 1 million 2014 CRC bins of reported revenue, comparing the periods before (2016–2018) and after the reform (2021–2023). Panel A presents these changes around the first reported revenue threshold, while Panel B shows those around the second reported revenue threshold.

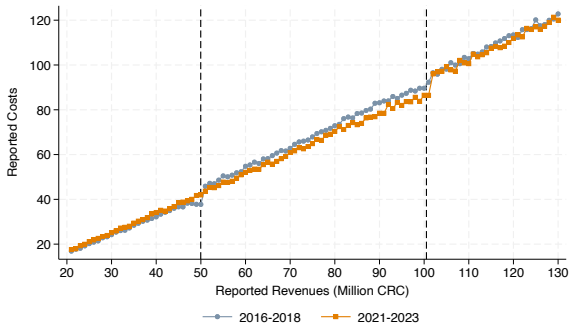
Figure 4. Firm Density, and Average Profit Margin, Costs and Employment, Before and After the Reform



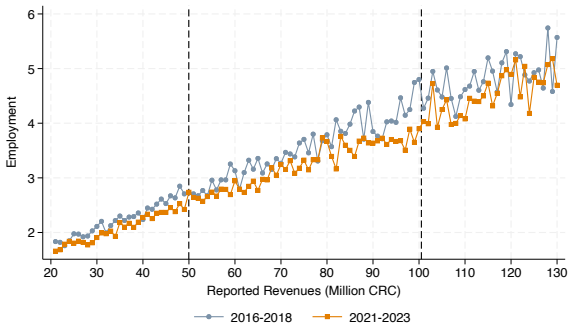
A. Firm density



B. Average profit margin



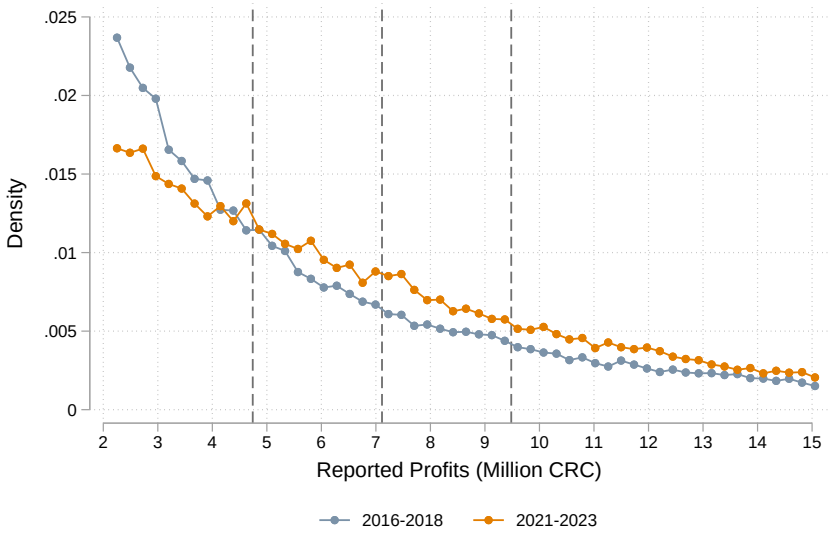
C. Average costs



D. Average employment

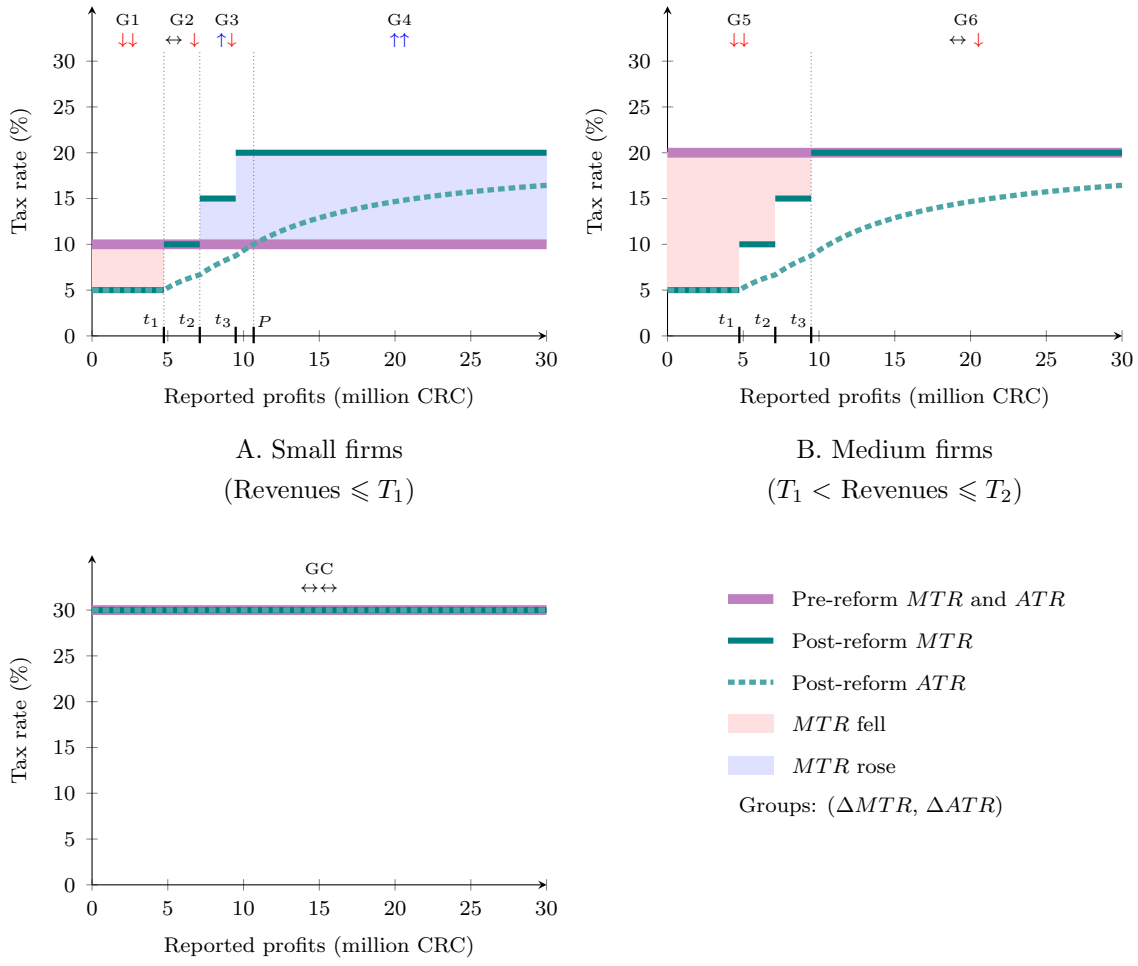
Notes: This figure shows the key patterns of the firms' data before and after the reform, for the years 2016–2018 in blue and 2021–2023 in orange, respectively. Panel A displays firm density by reported revenue bin. Panel B shows average profit margin by reported revenue bin, where profit margin is defined as profit over revenue. Panel C shows average reported costs by reported revenue bin. Panel D illustrates average number of employees registered by the firms to the Social Security Authority by reported revenue bin. Bin size is 1 million 2014 CRC.

Figure 6. Firm Density by Reported Profits, Before and After the Reform



Notes: This figure shows the density of reported profits of firms with reported revenue up to the second threshold T_2, before (2016–2018) and after the reform (2021–2023). Bin size is 0.24 million 2014 CRC. Dashed lines show the new profit thresholds t_1, t_2 and t_3.

Figure 7. Marginal and Average Tax Rates, Before and After the Reform



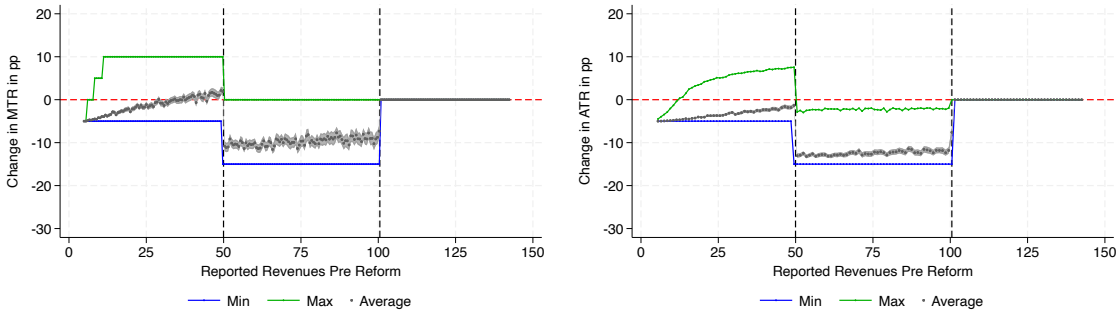
A. Small firms
(Revenues $\leqslant T_1$)

B. Medium firms
($T_1 <$ Revenues $\leqslant T_2$)

C. Large firms
(Revenues $> T_2$)

Notes: This figure illustrates the marginal and average tax rates faced by firms before and after the reform, for firms reporting the same revenue and profits in both periods. Panel A shows that small firms faced a marginal and average tax rate of 10% before the reform. After the reform, their marginal tax rate increases in steps from 5% to 20% with reported profits, and their average tax rate increases smoothly from 5% towards 20%. Panel B shows that medium firms faced a rate of 20% before the reform and the same post-reform schedule after it. Panel C shows that large firms faced a marginal and average tax rate of 30% before and after the reform. Shaded areas mark the profit ranges in which the marginal tax rate fell (red) or rose (blue). Firms are classified into seven groups by firm size and reported profits, depending on how the reform changed their marginal and average tax rates for an unchanged tax report, shown by the arrows above each panel (↓ fell, ↑ rose, ↔ unchanged). Dotted lines mark the boundaries between groups. t_1, t_2 and t_3 are the post-reform profit thresholds, and P is the level of profits at which the change in the average tax rate of small firms turns from negative to positive. Groups G1 to G6 are treatment groups, since at least their average tax rate changed; GC is the control group, since neither rate changed.

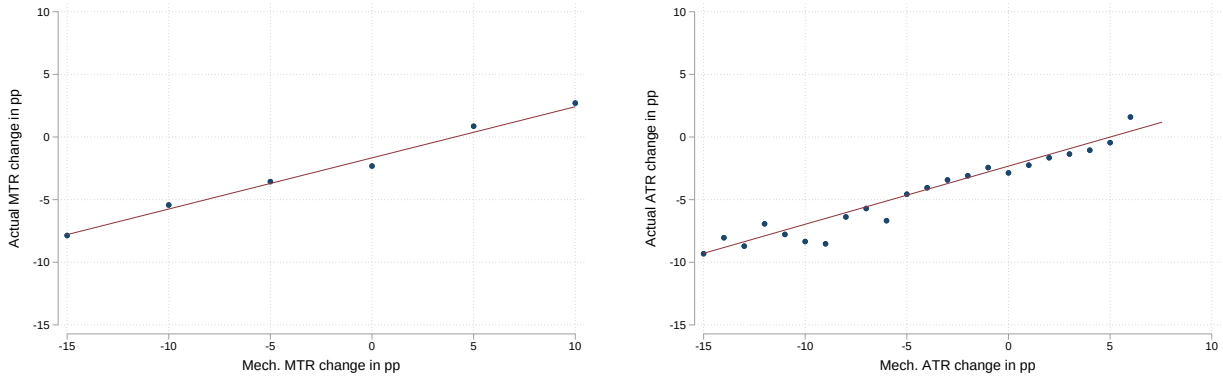
Figure 8. Reform-Induced Mechanical Marginal and Average Tax Rate Changes



A. Marginal tax rate B. Average tax rate

Notes: Panel A illustrates the average, minimum, and maximum mechanical marginal tax rate changes, while Panel B illustrates the average, minimum, and maximum mechanical average tax rate changes induced by the reform. Both are calculated based on firms' average reported revenue and profits from 2016–2019, by 1 million 2014 CRC reported revenue bins.

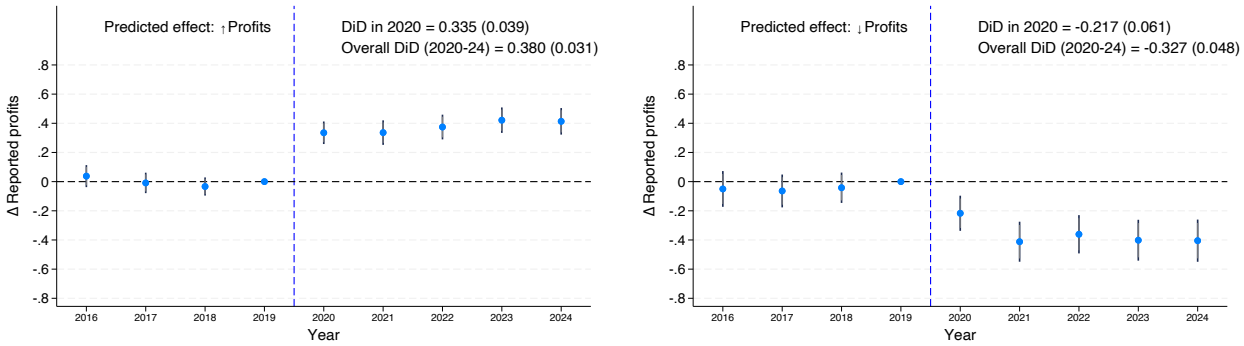
Figure 9. Mechanical and Actual Tax Rate Changes, 2020



A. Marginal tax rate B. Average tax rate

Notes: This figure compares the mechanical and actual changes in the marginal and average tax rates of treated firms (Groups G1 to G6) in our preferred balanced-panel sample. The mechanical change is calculated from 2016–2019 average revenue and reported profits. The actual change is the rate on the firm's 2020 reports under the new schedule, minus its average actual rate in 2016–2019. Dots show average actual changes by mechanical change; the line is a linear fit across firms.

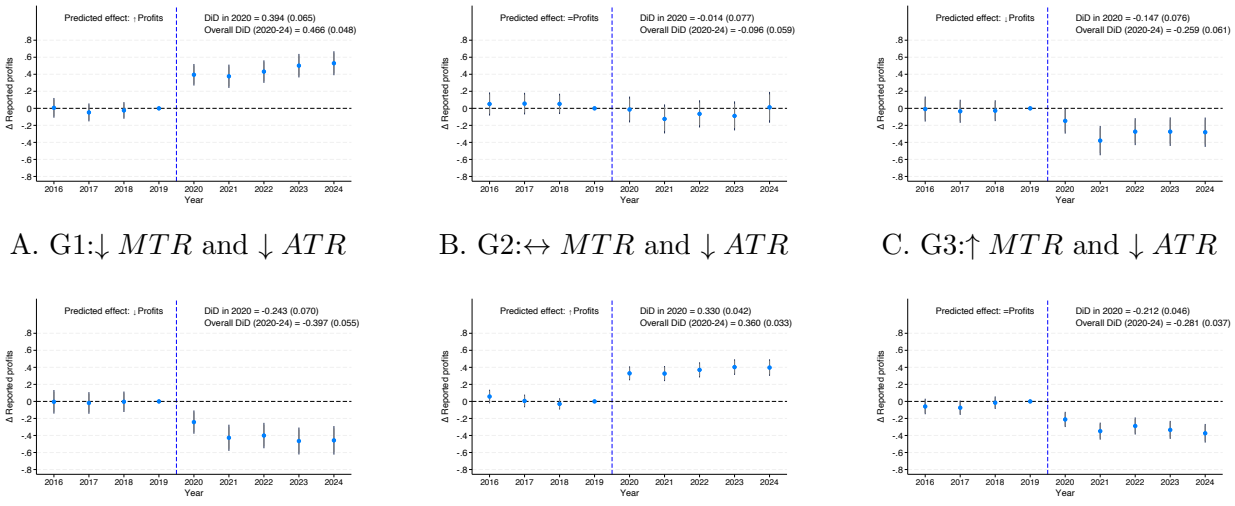
Figure 10. Difference-in-Differences by MTR Treatment Groups



A. Effects on treated group $\downarrow MTR$ B. Effects on treated group $\uparrow MTR$

Notes: This figure presents the 2016–2024 difference-in-differences event study estimates of the dynamic effects of the reform on reported profits for treatment groups that experienced changes in their marginal tax rates, compared to the control group of large firms (GC), using our preferred balanced-panel sample. Treatment is determined using the mechanical marginal tax rate change, calculated from 2016–2019 average revenue and reported profits. A Poisson pseudo-maximum likelihood model is employed to account for zeros in the dependent variable. All estimations incorporate firm and year fixed effects, as well as year fixed effects interacted with pre-reform industry, employment deciles and wage bill-to-revenue ratio deciles. Panel A displays the effects for firms that experienced a decrease in the marginal tax rate, while Panel B shows the effects for those facing an increase. Standard errors are clustered at the firm level.

Figure 11. Difference-in-Differences by MTR and ATR Treatment Groups



A. G1:↓ MTR and ↓ ATR B. G2:↔ MTR and ↓ ATR C. G3:↑ MTR and ↓ ATR

D. G4:↑ MTR and ↑ ATR E. G5:↓ MTR and ↓ ATR F. G6:↔ MTR and ↓ ATR

Notes: This figure presents the 2016–2024 difference-in-differences event study estimates of the dynamic effects of the reform on reported profits for the six treatment groups, defined based on changes in marginal and average tax rates, relative to the control group of large firms (GC), using our preferred balanced-panel sample. Treatment is determined using the mechanical changes in marginal and average tax rates, calculated from 2016–2019 average revenue and reported profits. A Poisson pseudo-maximum likelihood model is employed to account for zeros in the dependent variable. All estimations incorporate firm and year fixed effects, as well as year fixed effects interacted with pre-reform industry, employment deciles and wage bill-to-revenue ratio deciles. Panel A shows the effects for Group 1, which consists of small firms that experienced a decrease in both their marginal and average tax rates. Panel B displays the effects for Group 2, including small firms that experienced no change in their marginal tax rate but a decrease in their average tax rate. Panel C illustrates the effects for Group 3, comprising firms that experienced an increase in their marginal tax rate and a decrease in their average tax rate. Panel D presents the effects for Group 4, featuring firms that experienced an increase in both their marginal and average tax rates. Panel E presents the effects for Group 5, consisting of medium-sized firms that experienced a decrease in both their marginal and average tax rates. Panel F shows the effects for Group 6, comprising medium-sized firms that experienced no change in their marginal tax rate but a decrease in their average tax rate. Standard errors are clustered at the firm level.

8 Appendix

Table A1. Pre-Reform Characteristics by Firm Size and Industry

	Small	Medium	Large	Total
Panel A: Main sample (balanced panel, employment records)				
Revenue (million CRC)	29.8	80.9	146.2	68.1
	(14.2)	(15.9)	(23.7)	(47.3)
Costs (million CRC)	27.7	75.5	139.3	64.1
	(18.8)	(22.4)	(27.2)	(47.5)
Profit margin	0.16	0.10	0.06	0.12
	(0.15)	(0.11)	(0.07)	(0.13)
Employment (employees)	1.6	3.1	4.9	2.7
	(1.5)	(2.6)	(3.7)	(2.7)
Wage bill (million CRC)	5.9	12.8	21.6	11.1
	(6.8)	(12.0)	(17.7)	(12.8)
Industry (% of firms)				
Agriculture, forestry & fishing	8.6	6.3	6.5	7.5
Manufacturing, mining & other industries	6.0	8.9	9.7	7.6
Construction	4.0	5.3	4.8	4.5
Wholesale, retail, transportation & storage	20.1	26.0	34.1	24.7
Accommodation & food services	3.8	4.5	5.8	4.4
Information & communication	2.3	2.7	2.8	2.5
Financial & insurance	1.4	1.4	1.2	1.4
Real estate	15.5	9.8	5.4	11.7
Prof, scientific, technical & admin	13.9	14.7	13.2	14.0
Education, human health & social work	16.7	12.0	10.9	14.1
Other services	7.7	8.4	5.5	7.5
No industry recorded	0.0	0.1	0.1	0.1
Firms	6,264	3,973	2,431	12,668
Firm-years	56,376	35,757	21,879	114,012
Panel B: All firms (unbalanced panel)				
Revenue (million CRC)	23.7	79.4	145.2	48.6
	(14.6)	(15.8)	(24.1)	(43.4)
Costs (million CRC)	21.9	74.7	138.4	45.7
	(18.3)	(23.3)	(29.4)	(43.8)
Profit margin	0.17	0.10	0.06	0.14
	(0.19)	(0.12)	(0.07)	(0.17)
Industry (% of firms)				
Agriculture, forestry & fishing	6.9	5.3	5.8	6.4
Manufacturing, mining & other industries	5.9	7.4	7.9	6.4
Construction	5.2	6.9	5.4	5.6
Wholesale, retail, transportation & storage	20.4	28.1	34.2	23.6
Accommodation & food services	3.5	4.7	6.7	4.1
Information & communication	3.4	2.9	2.5	3.2
Financial & insurance	2.1	1.4	1.4	1.9
Real estate	8.8	6.8	4.6	7.9
Prof, scientific, technical & admin	19.0	15.6	13.1	17.6
Education, human health & social work	12.2	11.2	11.4	11.9
Other services	8.3	8.0	6.0	8.0
No industry recorded	4.2	1.5	1.1	3.3
Firms	33,783	10,718	5,326	49,827
Firm-years	228,519	79,498	40,323	348,340

Notes: This table presents pre-reform characteristics by firm size and industry, for our main sample (Panel A) and for all firms in the unbalanced panel (Panel B), which correspond to Columns 4 and 1 of Table 1. Firm size is defined by pre-reform average revenue relative to the two revenue thresholds of the pre-reform schedule. The table reports means of each firm's 2016–2019 averages, with standard deviations across firms in parentheses, and the share of firms in each industry. Values are in million 2014 CRC and winsorised at the 1st and 99th percentiles within each year. Industry follows the eleven-category classification used as a control, with firms whose industry is not recorded shown separately.

Table A2. Heterogeneity by Firm Size: Marginal Tax Rate Effects on Reported Profits

	(1) ↓ MTR + ↑ MTR	(2) ↓ MTR	(3) ↑ MTR	(4) ↓ MTR + ↑ MTR	(5) ↓ MTR	(6) ↑ MTR	(7) ↓ MTR + ↑ MTR	(8) ↓ MTR	(9) ↑ MTR
$T \times$ Post	0.3201***	0.5967***	−0.3888***	0.2179***	0.4739***	−0.5294***	0.2728***	0.4636***	−0.4169***
	(0.0209)	(0.0211)	(0.0321)	(0.0392)	(0.0399)	(0.0762)	(0.0548)	(0.0583)	(0.0878)
× Larger small	−0.4217***	−0.0209	−0.0428	−0.3446***	−0.0005	0.1057	−0.3229***	−0.0047	0.0941
	(0.0183)	(0.0224)	(0.0318)	(0.0397)	(0.0459)	(0.0779)	(0.0406)	(0.0475)	(0.0749)
× Smaller medium	0.0685***	−0.2067***	–	0.1577***	−0.0983**	–	0.0887*	−0.1042**	–
	(0.0230)	(0.0232)		(0.0440)	(0.0446)		(0.0499)	(0.0522)	
× Larger medium	0.0990***	−0.1764***	–	0.1702***	−0.0857*	–	0.0862	−0.1042*	–
	(0.0269)	(0.0270)		(0.0464)	(0.0470)		(0.0566)	(0.0593)	
Employment sample	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Balanced panel	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes
N	300,833	266,019	74,558	95,931	85,320	32,292	95,931	85,320	32,292

Notes: This table presents triple-difference estimates of the effects of changes in the marginal tax rate on reported profits by pre-reform firm size, relative to the control group of large firms (GC). Treatment is determined using the mechanical marginal tax rate change, calculated from 2016–2019 average revenue and reported profits; Groups G2 and G6 are excluded. Columns 1–3 use the unbalanced panel of all firms and Columns 4–9 the balanced panel of firms with employment records. A Poisson pseudo-maximum likelihood model is employed to account for zeros in the dependent variable. All estimations incorporate firm and year fixed effects; additional controls are year fixed effects interacted with pre-reform industry, employment deciles and wage bill-to-revenue ratio deciles. Size groups are defined by pre-reform average revenue, cut at $T_1/2$, T_1, $T_1 + (T_2 - T_1)/2$ and T_2; smaller small firms are the omitted category, so $T \times$ Post is the effect for that group and each interaction is the difference from it. N includes the control group. Standard errors (in parentheses) are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A3. Heterogeneity by Industry: Marginal Tax Rate Effects on Reported Profits

	(1) ↓ MTR + ↑ MTR	(2) ↓ MTR	(3) ↑ MTR	(4) ↓ MTR + ↑ MTR	(5) ↓ MTR	(6) ↑ MTR	(7) ↓ MTR + ↑ MTR	(8) ↓ MTR	(9) ↑ MTR
$T \times$ Post	0.0679**	0.3450***	−0.2525***	0.0120	0.1936***	−0.3015***	0.2150***	0.2832***	−0.1463
	(0.0267)	(0.0311)	(0.0361)	(0.0432)	(0.0488)	(0.0674)	(0.0683)	(0.0715)	(0.0903)
× Agriculture, forestry & fishing	0.2325***	0.2864***	−0.1842**	0.2611***	0.4127***	−0.1458	−0.1468	0.0290	−0.5094***
	(0.0421)	(0.0441)	(0.0877)	(0.0701)	(0.0700)	(0.1540)	(0.1199)	(0.1209)	(0.1762)
× Manufacturing, mining & other industries	0.2319***	0.1627***	−0.2510***	0.3034***	0.2797***	−0.2149	0.0380	0.0569	−0.4241***
	(0.0377)	(0.0407)	(0.0775)	(0.0638)	(0.0668)	(0.1332)	(0.1075)	(0.1090)	(0.1632)
× Construction	0.2238***	0.1817***	−0.0932	0.3761***	0.4218***	0.0933	0.2680**	0.3419**	0.0281
	(0.0405)	(0.0445)	(0.0763)	(0.0778)	(0.0851)	(0.1308)	(0.1287)	(0.1346)	(0.1714)
× Wholesale, retail, transportation & storage	0.2675***	0.2264***	−0.1399***	0.2580***	0.2750***	−0.0122	−0.0090	0.0355	−0.2458**
	(0.0274)	(0.0316)	(0.0509)	(0.0481)	(0.0537)	(0.0896)	(0.0807)	(0.0835)	(0.1105)
× Accommodation & food services	0.1971***	0.1504***	−0.1531	0.1191	0.1228	−0.1257	0.0811	0.1195	−0.0966
	(0.0466)	(0.0499)	(0.0958)	(0.0778)	(0.0854)	(0.1455)	(0.1245)	(0.1303)	(0.1759)
× Information & communication	0.2572***	0.3502***	−0.1105	0.3931***	0.4214***	0.1069	0.2673*	0.3068*	−0.0087
	(0.0482)	(0.0533)	(0.0707)	(0.0958)	(0.1064)	(0.1427)	(0.1591)	(0.1657)	(0.1922)
× Financial & insurance	0.1369*	0.8689***	−0.3030***	0.3071*	0.9685***	−0.2527*	0.1766	0.8201***	−0.2507
	(0.0750)	(0.1060)	(0.0770)	(0.1646)	(0.2182)	(0.1337)	(0.2375)	(0.2810)	(0.2397)
× Real estate	−0.1931***	0.1406***	−0.2786***	−0.2328***	0.1852**	−0.2557***	−0.0696	0.3172***	−0.0724
	(0.0345)	(0.0461)	(0.0465)	(0.0538)	(0.0740)	(0.0777)	(0.1073)	(0.1173)	(0.1238)
× Prof, scientific, technical & admin	−0.0084	0.1304***	−0.1643***	0.0441	0.1781***	−0.2139**	−0.0595	0.0920	−0.2832**
	(0.0271)	(0.0325)	(0.0410)	(0.0517)	(0.0571)	(0.0881)	(0.0945)	(0.0976)	(0.1176)
× Other services	−0.0062	0.0149	−0.3469***	0.1032*	0.0993	−0.1641	−0.1171	−0.1083	−0.3480**
	(0.0334)	(0.0385)	(0.0587)	(0.0626)	(0.0694)	(0.1250)	(0.1213)	(0.1254)	(0.1620)
Employment sample	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Balanced panel	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes
N	300,833	266,019	74,558	95,931	85,320	32,292	95,931	85,320	32,292

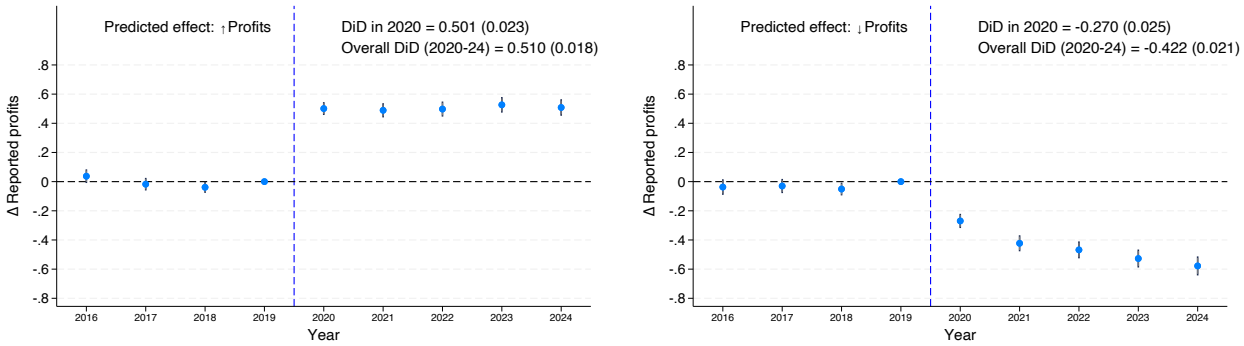
Notes: This table presents triple-difference estimates of the effects of changes in the marginal tax rate on reported profits by industry, relative to the control group of large firms (GC). Treatment is determined using the mechanical marginal tax rate change, calculated from 2016–2019 average revenue and reported profits; Groups G2 and G6 are excluded. Columns 1–3 use the unbalanced panel of all firms and Columns 4–9 the balanced panel of firms with employment records. A Poisson pseudo-maximum likelihood model is employed to account for zeros in the dependent variable. All estimations incorporate firm and year fixed effects; additional controls are year fixed effects interacted with pre-reform industry, employment deciles and wage bill-to-revenue ratio deciles. Education, human health and social work is the omitted category, so $T \times$ Post is the effect for that category and each interaction is the difference from it. Firms whose industry is not recorded are grouped with other services. N includes the control group. Standard errors (in parentheses) are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A4. GDP 2019 and GDP Growth 2020, by Industry

	GDP 2019	GDP Growth 2020
Agriculture, forestry & fishing	1,654,172	-1.4
Manufacturing, mining & other industries	5,335,924	2.9
Construction	1,454,936	-0.9
Wholesale, retail, transportation & storage	4,869,581	**-10.8**
Accommodation & food services	1,208,238	**-36.3**
Information & communication	1,601,438	1.8
Financial & insurance	2,028,405	5.6
Real estate	2,938,306	-2.3
Prof, scientific, technical & admin	4,530,372	-3.2
Education, human health & social work	5,032,630	-1.8
Other services & NA	1,313,058	**-18.1**

Notes: This table presents Costa Rica's 2019 GDP and 2020 GDP growth rates, both by industry, used to define the industries most affected by the COVID-19 shock. Industries with a GDP contraction in 2020 larger than 10%—specifically accommodation & food services; other services & NA; and wholesale, retail, transportation & storage— are classified as the most affected industries (in bold). This classification is used in the triple-difference specification to test the robustness of our main results to potential COVID-19 confounding effects.

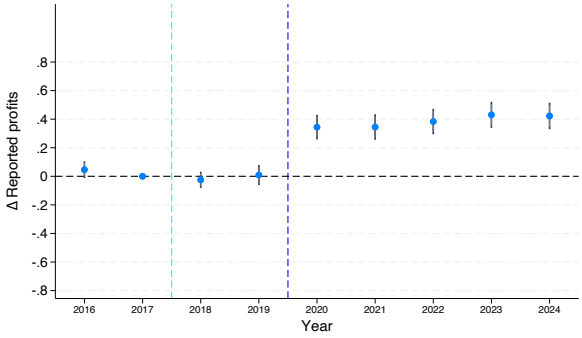
Figure A1. Difference-in-Differences by MTR Treatment Groups, Unbalanced-Panel Sample



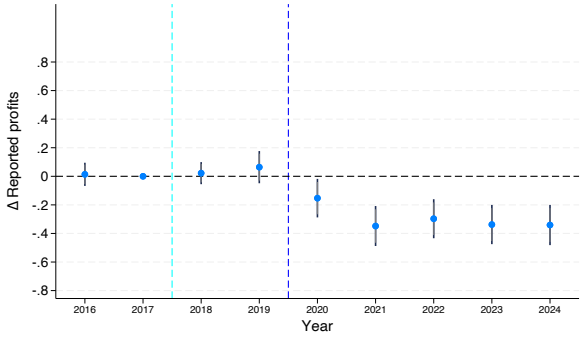
A. Effect on treated group $\downarrow MTR$ B. Effect on treated group $\uparrow MTR$

Notes: This figure presents the 2016–2024 difference-in-differences event study estimates of the dynamic effects of the reform on reported profits for treatment groups that experienced changes in their marginal tax rates, compared to the control group of large firms (GC), using the unbalanced panel of all firms. Treatment is determined using the mechanical marginal tax rate change, calculated from 2016–2019 average revenue and reported profits. A Poisson pseudo-maximum likelihood model is employed to account for zeros in the dependent variable. All estimations incorporate firm and year fixed effects, without additional controls. Panel A displays the effects for firms that experienced a decrease in the marginal tax rate, while Panel B shows the effects for those facing an increase. Standard errors are clustered at the firm level.

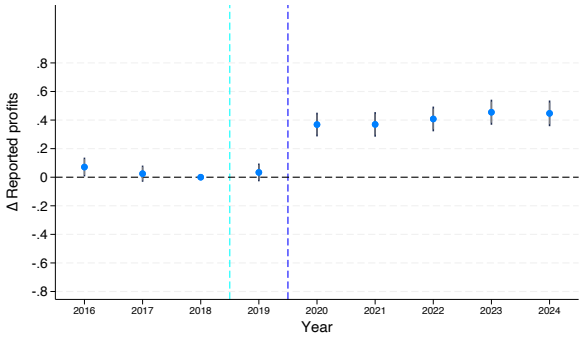
Figure A2. Difference-in-Differences by MTR Treatment Groups, Change in Treatment Year Placebo Tests



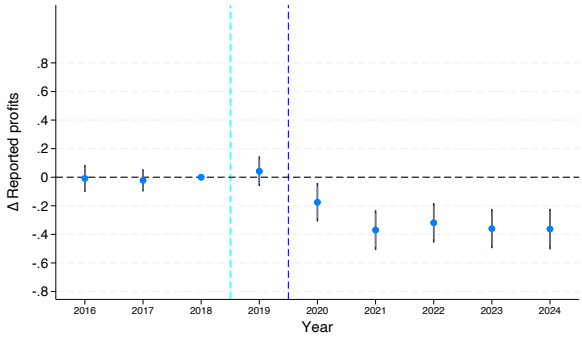
A. Effects on treated group ↓ MTR, placebo 2018



B. Effects on treated group ↑ MTR, placebo 2018



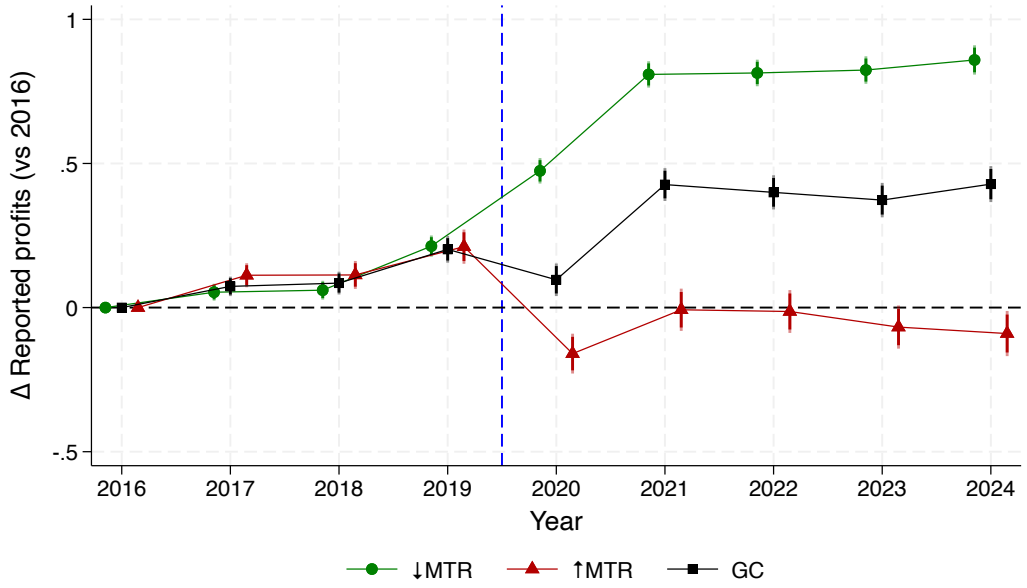
C. Effects on treated group ↓ MTR, placebo 2019



D. Effects on treated group ↑ MTR, placebo 2019

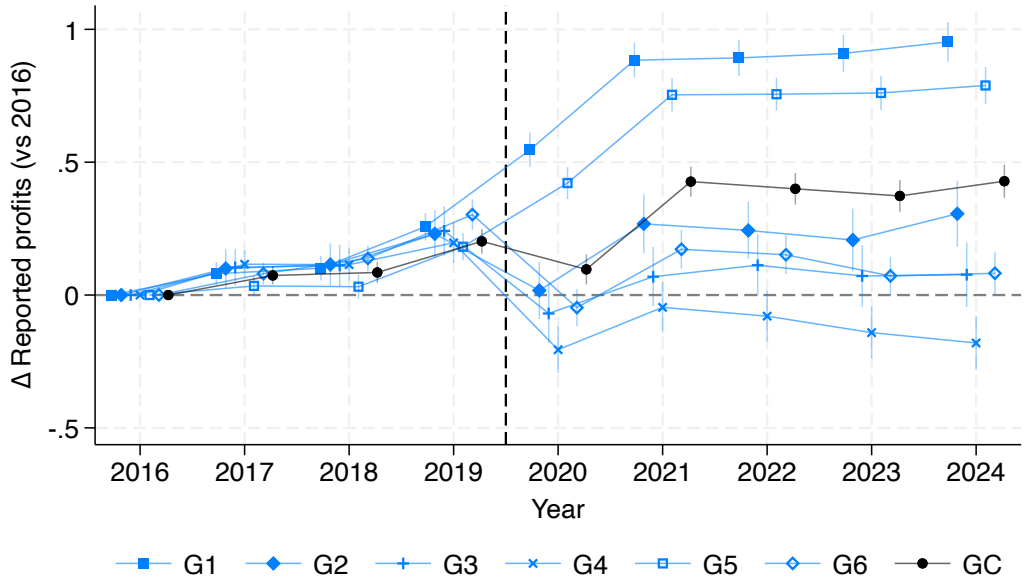
Notes: This figure presents placebo event study estimations to test the no-anticipation assumption in our difference-in-differences strategy. We replicate the preferred event study specification, setting 2018 and 2019 as hypothetical treatment years instead of 2020 (light dashed lines); the base year is 2017 in Panels A and B and 2018 in Panels C and D. Treatment groups correspond to firms that experienced a decrease or an increase in their mechanical marginal tax rate, relative to the control group of large firms (GC). Standard errors are clustered at the firm level.

Figure A3. Raw Event Studies of Reported Profits, by MTR Treatment Group



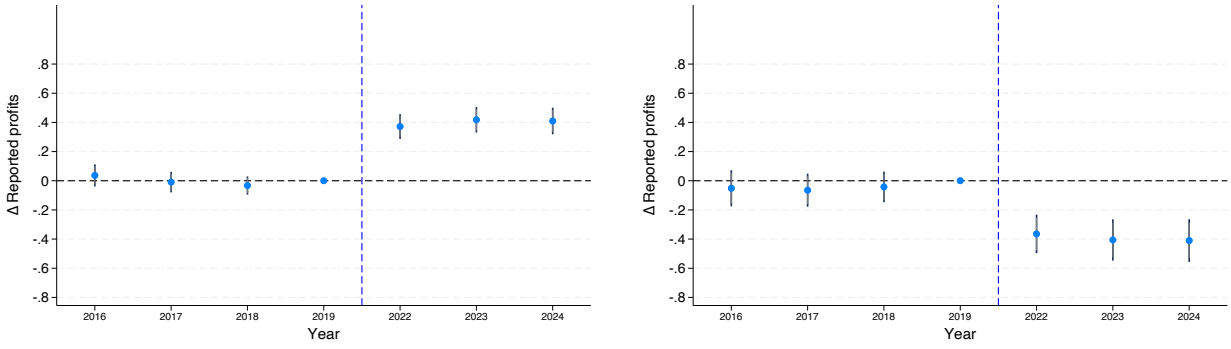
Notes: This figure presents raw event studies of reported profits for 2016–2024, separately for firms that experienced a decrease in their marginal tax rate (Groups G1 and G5), firms that experienced an increase (Groups G3 and G4), and the control group of large firms (GC). Groups are determined using the mechanical marginal tax rate change, calculated from 2016–2019 average revenue and reported profits. For each group, we estimate a Poisson pseudo-maximum likelihood model on that group's firms only, with firm and year fixed effects and no further controls, using 2016 as the base year. In our preferred balanced-panel sample of firms with employment records, each coefficient equals the log change in the group's mean reported profits relative to 2016. Bars show 95% confidence intervals. Standard errors are clustered at the firm level.

Figure A4. Raw Event Studies of Reported Profits, by Group



Notes: This figure presents raw event studies of reported profits for 2016–2024 for each of the six treatment groups and the control group of large firms (GC), estimated as in Figure A3: a Poisson pseudo-maximum likelihood model on each group's firms only, with firm and year fixed effects and 2016 as the base year. Bars show 95% confidence intervals. Standard errors are clustered at the firm level.

Figure A5. Difference-in-Differences by MTR Treatment Groups, Robustness Excluding COVID-19 Years

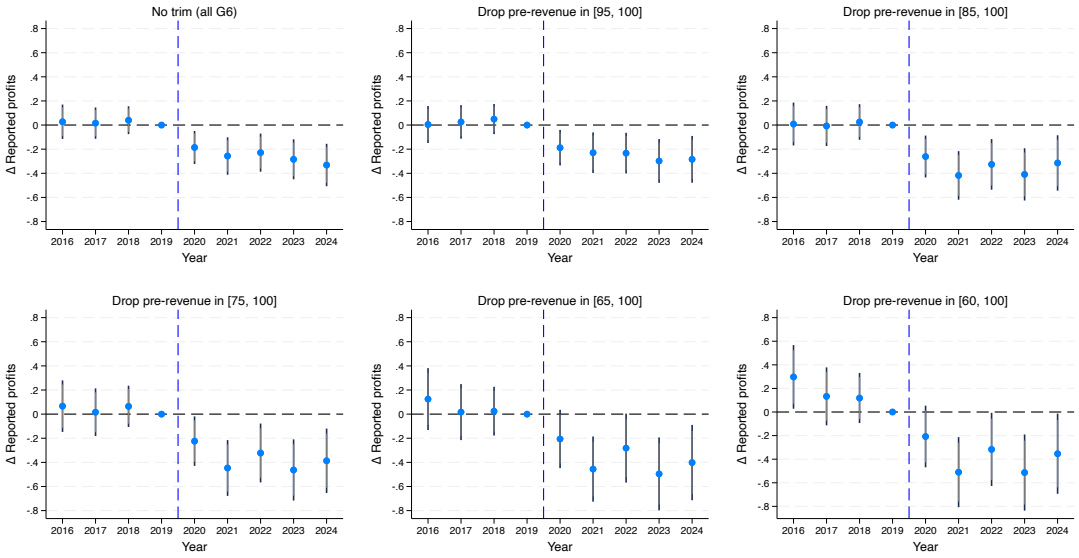


A. Effects on treated group $\downarrow MTR$

B. Effects on treated group $\uparrow MTR$

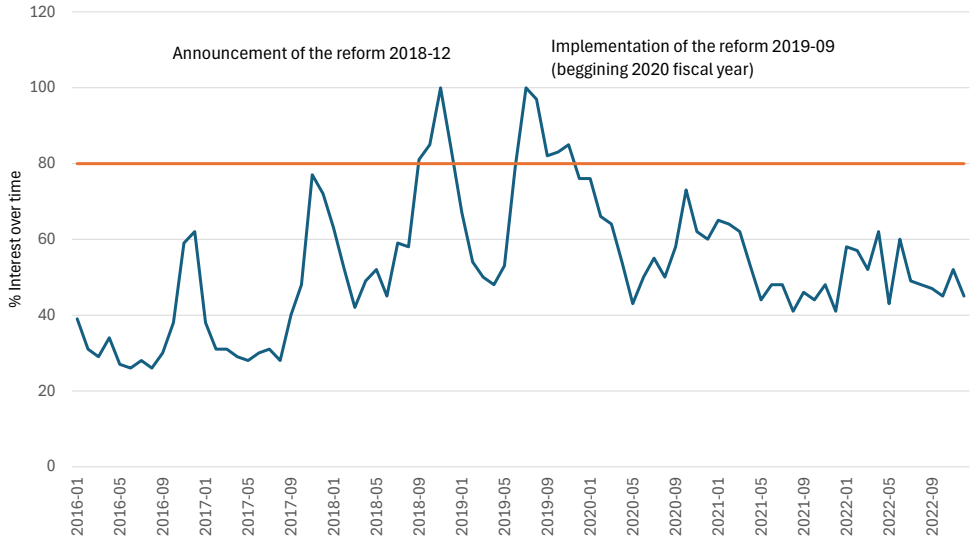
Notes: This figure presents the event study estimates of our preferred specification, excluding the 2020 and 2021 fiscal years, which were most affected by the COVID-19 pandemic. Treatment groups correspond to firms that experienced a decrease or an increase in their mechanical marginal tax rate, relative to the control group of large firms (GC). Standard errors are clustered at the firm level.

Figure A6. Effects on Reported Profits of Group G6, Excluding Firms Closest to the Second Revenue Threshold



Notes: This figure presents the event study estimates of the dynamic effects of the reform on reported profits for Group G6 (medium firms with no change in MTR and a decrease in ATR), relative to the control group of large firms (GC), using our preferred balanced-panel sample, with 2019 as the base year. Panel (a) uses all G6 firms (1,046 firms). Panels (b) to (f) progressively exclude G6 firms with the highest pre-reform revenue, those closest to the second revenue threshold T_2: firms with 2016–2019 average revenue above 95, 85, 75, 65 and 60 million 2014 CRC, respectively, keeping 959, 771, 559, 338 and 234 firms. Each sample is nested in the previous one, and the control group is the same in all panels. A Poisson pseudo-maximum likelihood model is employed to account for zeros in the dependent variable. All estimations incorporate firm and year fixed effects, and year fixed effects interacted with pre-reform industry, employment bins, wage bill-to-revenue ratio bins and log pre-reform revenue. Bars show 95% confidence intervals. Standard errors are clustered at the firm level.

Figure A7. Google Searches for "*Ministerio de Hacienda*" (Ministry of Finance)



Notes: This figure displays the percentage interest for the term "Ministerio de Hacienda" (Ministry of Finance) based on Google search data from 2016 to 2022, with 100% representing peak interest. Notable spikes in search interest coincide with the announcement of the tax reform in December 2018 and its implementation in October 2019.